

Who Speaks, Which Way, and When?

Simultaneous Speech Translation of Two-Language Conversations

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Abstract

Simultaneous speech translation is mostly built and evaluated for one speaker translated in one fixed direction. Machine interpreting between two people who share no language breaks both assumptions: one device hears both speakers in a single mono stream, and the system must decide in real time who is speaking, which way to translate, and when to commit text it can never retract. We formalise this task as conversational simultaneous speech-to-text translation (CST) and introduce CST-Bench, which renders two-language dialogues as continuous single-channel sessions with natural gaps, translation-mediated gaps or turn-end overlap; it pairs German–English telephone dialogues with real speech and synthetic timing (86 sessions, 640 turns with human translations), released as deterministic build scripts and checksums, with an evaluation-only synthetic track for English–German and Korean–English. Its session-level scorer reports per-direction translation quality, lag within a turn, the listener’s wait after each turn, switch latency and wrong-direction output, plus speaker-attributed transcription and diarization errors; controlled perturbations test each metric, and Kendall’s τ between system rankings on synthetic and real sessions is [TBD]. With realistic instead of oracle segmentation, the best open pipeline loses [TBD] COMET and emits [TBD]% wrong-direction words under turn-end overlap, and segmentation and routing account for [TBD] and [TBD] COMET points of this loss. We also introduce MODEL, an open streaming speech LLM that writes speaker-tagged transcripts and direction-tagged translations into one token stream on an 80 ms clock, with translation commits learned

from alignment-verified meaning units; without oracle input, it recovers [TBD]% of the gap between oracle and realistic segmentation at a median wait of [TBD] s after each turn. *[Draft: planned; the synthetic track, the overlap renders, most scorer modules and MODEL beyond its streaming ASR backbone are not built yet, and all results are pending]*

1. Introduction

Two people who share no language can talk through a device that translates both sides, a goal of early speech translation systems (Waibel, 1996; Lavie et al., 1996; Wahlster, 2000) that deployed services such as Skype Translator have since offered (Lewis, 2015). The device stands in for a dialogue interpreter, who coordinates turns as well as translating them (Wadensjö, 1998; Roy, 2000). Conversation leaves it little time: gaps between turns are of the order of 200 ms while producing language takes over 600 ms (Levinson & Torreira, 2015), average gaps lie within 250 ms of the cross-language mean in 10 languages (Stivers et al., 2009), and about 40% of between-speaker intervals are overlaps (Heldner & Edlund, 2010). Simultaneous speech translation (SimulST) has matured from wait- k and monotonic attention (Ma et al., 2019; 2020b;c) through local-agreement and attention-based policies, also for unbounded audio (Liu et al., 2020; Polák et al., 2022; Papi et al., 2023a; 2024), to large streaming models (Seamless Communication et al., 2023; Ouyang et al., 2025; Labiausse et al., 2025), and IWSLT now evaluates it on long unsegmented audio (Abdulummin et al., 2025; Adelani et al., 2026). Yet most work uses human pre-segmented speech (Papi et al., 2025), and SimulST benchmarks, segmented or not, translate one source language in one fixed direction, mostly from a single speaker.

Conversation changes the problem in four ways (Figure 1). (i) *The direction is latent*: one device hears both speakers in one mono stream, so the system must find, as it listens, each change of speaker and hence which way to translate. With one language per speaker, as in our scope, this is spoken language identification, but changes must be detected within gaps of a few hundred milliseconds or during over-

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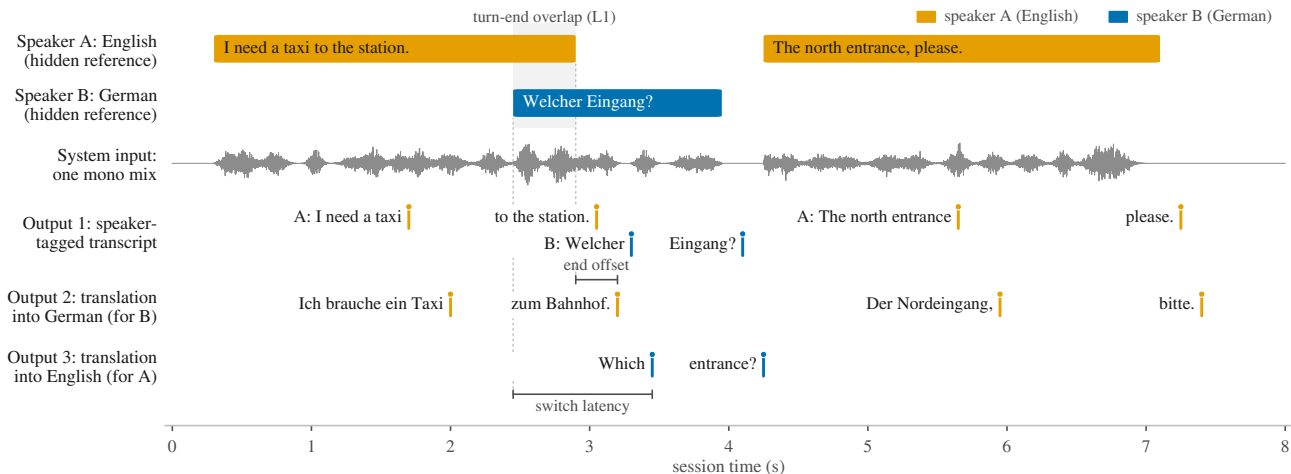


Figure 1. CST on one mono stream (illustrative; invented utterances). Speakers A (English) and B (German) share one device. The system hears only the mix and must emit, without retraction, translations into each listener’s language, choosing the direction itself, and may also emit a speaker-tagged transcript (Output 1). Ticks mark committed pieces at their emission time, coloured by the speaker whose speech they render. EndOffset (*end offset*): time from the end of a turn to the last word of its translation, i.e., how long the listener waits. Switch latency: time from the start of a turn in the new direction to the first word of its translation. Grey band: turn-end overlap (condition L1). Reference turns are hidden from the system and used only for scoring (Section 4 defines the metrics).

lap. (ii) *Turns are short and close*: the median turn in our real dialogues lasts 2.86 s after silence trimming, and in natural conversation a reply follows after a short gap or overlaps the end of the turn it answers. (iii) *Output is committed*: a listener may reply as soon as text appears, and a later revision cannot undo that reply, so we score append-only output; re-translation systems (Arivazhagan et al., 2020) take part through a commit rule such as local agreement (Section 6.1). (iv) *Latency is felt at turn ends*: the listener waits after each turn (EndOffset) and at each change of direction (switch latency), beyond the lag within a turn that StreamLAAL measures (Papi et al., 2024). We therefore pose conversational simultaneous speech-to-text translation (CST) as streaming structured prediction from a mixture, with a latent speaker and direction state and append-only, time-stamped emissions on several channels: a translation into each language and, optionally, a speaker-attributed transcript and speaker activity.

Prior systems cover parts of this setting (Table 1). JS-TAR (Moritz et al., 2025) jointly streams recognition and translation of bilingual conversations and also infers who speaks, wearer or partner, and hence the direction, but from spatial cues of multi-microphone smart glasses and on in-house data. DiariST (Yang et al., 2024) adds diarization to streaming translation of multi-speaker meetings, but only from Chinese to English. STAC-ST (Zuluaga-Gomez et al., 2023) handles two speakers in one channel with speaker-turn tokens, but offline and from one source language. Seed LiveInterpret 2.0 (Cheng et al., 2025) names multi-speaker confusion as a challenge and depicts a bilingual conversation, but this closed system is evaluated per direction with-

out overlap and covers only Chinese–English, which CST-Bench lacks, so we do not run it. We claim neither single-channel input nor conversational translation as new. To our knowledge, however, no public benchmark evaluates streaming translation of both sides of a two-language conversation from one mono stream, where speaker changes, directions and turn ends are latent and turns may overlap, and no open baselines or models exist for this setting.

We ask (RQ1) how much of the difficulty lies in segmentation and routing (choosing each turn’s direction) rather than translation; (RQ2) how quality, lag within a turn and listener wait trade off as turn-taking tightens from mediated to natural gaps and turn-end overlap, and for a distant language pair; (RQ3) whether one streaming model that writes transcripts and translations into one token stream closes the gap between oracle and realistic segmentation without oracle input; and (RQ4) whether synthetic sessions rank systems as real ones do. Our contributions are:

- **Task and protocol** (Section 3): CST with irrevocable, time-stamped outputs whose language encodes the direction, scored under oracle and realistic segmentation.
- **CST-Bench** (Section 4): 86 German–English telephone dialogues (CST-Bench-TAXI) with real speech, 640 human-translated turns and rendered timing (Bavarian Archive for Speech Signals (BAS), 2001), shipped as deterministic build scripts and checksums, and an evaluation-only synthetic En↔De and Ko↔En track (CST-Bench-Syn); both are rendered with natural gaps (L0) and with turn-end overlap (L1), the real

track also with translation-mediated gaps (L0). [Draft: planned; CST-Bench-Syn and the L1 renders are not built yet]

- **A scorer and its validation protocol** (cstbench eval; Section 4) for per-direction quality, StreamLAAL, EndOffset, switch latency and wrong-direction output, plus cpWER and DER, tested by controlled perturbations. [Draft: partly planned; v0.1 computes BLEU, chrF, StreamLAAL, EndOffset, empty turns and wrong-language words given turn IDs; only two perturbation rows exist, as preliminary runs, and the untranslated-source row exposes a StreamLAAL artefact (Table 10)]
- **A controlled study** (Section 6) of strong open systems under oracle and realistic wrappers, with an oracle-component factorial over segmentation, routing and overlap. [Draft: planned; only offline oracle cascades have been run]
- **MODEL** (Section 5): an open streaming speech LLM that writes speaker-tagged transcripts and direction-tagged translations into one token stream on an 80 ms clock and commits translations at alignment-verified meaning units; ablations test whether coupling turn-taking and translation decisions helps. [Draft: planned; only the streaming ASR backbone is built]

Given oracle turns, an offline ASR→LLM cascade reaches 42.66[†] (En→De) and 28.11[†] (De→En) BLEU on CST-Bench-TAXI ([†]preliminary; to be re-run under the final protocol). Without oracle turns, the best open pipeline loses [TBD] COMET, segmentation and routing explain [TBD]% and [TBD]% of this loss, MODEL recovers [TBD]% of it, and Kendall’s τ between synthetic and real system rankings is [TBD].

Scope. We study two speakers, two languages and text output, under natural gaps, mediated gaps and turn-end overlap; interruptions and barge-in, more than two speakers, code-switching and speech output are left to future work.

2. Related Work

Simultaneous and streaming speech translation. Beyond fixed wait- k (Ma et al., 2019), learned policies use monotonic attention (Arivazhagan et al., 2019; Ma et al., 2020b) or commit at meaning units (Zhang et al., 2020); local agreement (Liu et al., 2020; Polák et al., 2022; Macháček et al., 2023) and attention (Papi et al., 2023a) run offline models simultaneously, also on unbounded audio (Papi et al., 2024) and with LLM translators (Fuxa & Macháček, 2026). LLM-based systems translate streaming speech (Ouyang et al., 2025; Guo et al., 2025; Cheng et al., 2024), and StreamSpeech, SeamlessStreaming and

Table 1. CST against the closest settings and systems. Spk×Lang: speakers × languages in one input stream; Auto dir.: the system chooses the direction of each turn; Ovl.: overlapping speech evaluated (sim.: simulated overlap); Str.: streaming; Out.: A transcript, T translation, D speaker labels or turns, S translated speech; Pub. data, Open wts: public evaluation data, open weights. Segm.: pre-segmented speech (e.g., MuST-C); long: unsegmented long-form audio (e.g., IWSLT 2025–2026). ✓ yes, ✗ no, ○ partial, n.r. not reported. ^aTAXI under its free scientific licence from BAS, rebuilt with our scripts and checksums; CST-Bench-Syn under CC licences.

Work	Input	Spk× Lang	Dir.	Auto dir.	Ovl.	Str.	Out.	Pub. data	Open wts
SimulST (segm.)	mono	1×1	X→Y	✗	✗	✓	T	✓	○
StreamST (long)	mono	1×1	X→Y	✗	✗	✓	T	✓	○
SeamlessStreaming	mono	1×1	X→Y	✗	✗	✓	T,S	✓	✓
StreamSpeech	mono	1×1	X→En	✗	✗	✓	A,T,S	✓	✓
Hibiki(-Zero)	mono	1×1	X→En	✗	✗	✓	T,S	✓	✓
STAC-ST	mono	2×1	Es→En	✗	✓	✗	A,T,D	✓	✗
DiariST	mono	N×1	Zh→En	✗	✓	✓	T,D	✓	✗
JSTAR	array	2×2	Es↔En	✓	sim.	✓	A,T,D	✗	✗
Seed LiveInterpret 2.0	n.r.	○	Zh↔En	○	✗	✓	T,S	✓	✗
CST-Bench + MODEL (ours)	mono	2×2	En↔De Ko↔En	✓	sim.	✓	A,T,D	✓ ^a	✓

[Draft: planned in our row: the L1 overlap renders, CST-Bench-Syn (incl. Ko↔En), transcript and diarization scoring, and MODEL with its weights; the TAXI build scripts and the translation scorer exist.]

Hibiki(-Zero) also output speech (Zhang et al., 2024; Seamless Communication et al., 2023; Labiausse et al., 2025; 2026). All of them are evaluated per direction and without speaker attribution, the setting that SimulST toolkits and latency metrics assume (Ma et al., 2020a; Gaido et al., 2025; Papi et al., 2022; Polák et al., 2025); several serve as our baselines (Section 6), and we reuse StreamLAAL, meaning units and Hibiki’s contextual alignment (Sections 4 and 5).

Joint transcription, translation and speakers. Transcripts and translations can share one model (Chen et al., 2021; Papi et al., 2023b), and serialized output training transcribes overlapping speakers with one decoder, also streaming in time order (t-SOT) and with speaker attribution (Kanda et al., 2020b; 2022a;b; 2020a; 2021); MODEL reuses this time-ordered serialization. Speaker embeddings let streaming ST detect speaker changes (Wang et al., 2025). Of the closest systems (Section 1), DiariST builds on t-SOT and scores speaker-attributed BLEU on meetings translated from Chinese (Yang et al., 2024); STAC-ST obtains its two-speaker channel by merging the two sides of Fisher-CALLHOME calls (Zuluaga-Gomez et al., 2023); JSTAR uses a fast-slow cascaded transducer encoder (Moritz et al., 2025); and Seed LiveInterpret 2.0 (Cheng et al., 2025) is evaluated on RealSI clips cut from public videos (Cheng et al., 2024). Table 1 contrasts their settings with ours; none of the four covers German or Korean.

Conversational translation systems and data. Conversational speech translation was studied as early as JANUS-II, which compared push-to-talk with cross-talk recording (Lavie et al., 1996), but public test data still fall short (Table 8): they are one-sided (MSLT; Federmann & Lewis, 2016; 2017), re-enacted (DRAL; Ward et al., 2023; 2024; Avila & Ward, 2023), mediated by machine translation and recorded per participant (InCroMin; Čechovič et al., 2025), from one source language (Fisher-CALLHOME; Post et al., 2013) or text-only (chat translation; Farajian et al., 2020; Liang et al., 2021). Code-switched speech alternates languages within, not between, speakers (Weller et al., 2022; Alastruey et al., 2023; Zhou et al., 2025). CST-Bench therefore builds on TAXI (Bavarian Archive for Speech Signals (BAS), 2001; Reichel et al., 2016), real German-English telephone dialogues with human translations of both sides (Section 4).

Diarization, turn-taking and multistream models. Neural diarization handles overlap, unknown speaker counts and streaming input (Fujita et al., 2019; Horiguchi et al., 2020; Xue et al., 2021); Sortformer orders speakers by arrival, also online (Park et al., 2025; Medennikov et al., 2025); transducers emit speaker roles or turns (El Shafey et al., 2019; Xia et al., 2022). Turn-taking models predict both speakers’ voice activity (Ekstedt & Skantze, 2022) or turn shifts from text (Ekstedt & Skantze, 2020) but learn from monolingual dyads, even when multilingual (Inoue et al., 2024). Full-duplex dialogue models (Nguyen et al., 2023; Défossez et al., 2024; Veluri et al., 2024) and delayed or step-synchronous decoders (Zeghidour et al., 2025; Seide et al., 2024) run on a fixed clock. Our diarization baselines are pyannote (Bredin, 2023) and Streaming Sortformer (Section 6.5), and MODEL adopts arrival-order speakers and one clock-synchronous token stream (Section 5). *[Draft: planned: the baseline runs, and MODEL beyond its streaming ASR backbone]*

3. Conversational Simultaneous Speech-to-Text Translation

We define CST independently of any architecture (Figure 1), so that all systems are scored by the same rules.

Definition 1 (CST). A *session* is a single-channel 16 kHz signal x on $[0, T]$ mixing two speakers, A and B , who share one dialogue context and speak languages $\ell_A \neq \ell_B$. Unobserved by the system, it consists of turns $n = 1, \dots, N$ with speaker $s_n \in \{A, B\}$, interval $[b_n, e_n]$ (turns may overlap), a transcript in ℓ_{s_n} and a reference translation r_n into the other language. Given the pair $\{\ell_A, \ell_B\}$ but no speaker enrolment, a CST system reads x as a stream and emits an append-only sequence of committed *pieces* (t_i, λ_i, y_i) : emission time, output language $\lambda_i \in \{\ell_A, \ell_B\}$, which selects a channel and thus the direction, and text. Optional

auxiliary channels carry a speaker-attributed transcript, as pieces $(t_j, \hat{s}_j, z_j)$ with speakers numbered by arrival, and two-speaker activity $\hat{a}(t) \in \{0, 1\}^2$. Every output depends only on x up to its emission time and is never revised.

Observability and clocks. A system thus decides who is speaking, which way to translate (λ_i) and when to commit (t_i). The pair is given, as a deployed device is configured for one pair; speakers, turn boundaries and directions are latent. A system may re-translate internally (Arivazhagan et al., 2020) as long as it only appends committed pieces. The ideal clock sets t_i to the audio consumed at commitment; the computation-aware clock adds processing time (Ma et al., 2020c).

Protocols. The *realistic* protocol, our main one, gives a system only x and the pair. The *oracle-segmentation* protocol, a diagnostic upper bound for single-direction systems, adds gold turn intervals and source languages, cutting each turn from the mix for the system that translates into the other language. Both use the same resegmentation (Section 4.5); scoring by gold turn identity is a diagnostic only.

Timing conditions. Following Heldner & Edlund (2010), gaps and pauses are silences between and within speakers, and an overlap is simultaneous speech at a speaker change. CST-Bench renders the same turns with everyday gaps (L0 natural), with 1–3 s gaps as when listeners first read the translation (L0 mediated), or with turn-end overlap (L1; Section 4.4). Turn timing ignores system output, so under L0 natural and L1, listeners reply as if translation were instantaneous. L1 overlaps at every speaker change, whereas about 40% of between-speaker intervals overlap in real conversation (Heldner & Edlund, 2010), so it is a stress condition. Content-free backchannels (L1b), which must neither be translated nor flip the direction, are at most an optional stress test; interruptions and barge-in are out of scope. *[Draft: L1 is planned; whether L1b enters the first release is undecided]*

CST as sequential prediction. At each step k , a system commits translation and transcript tokens Y_k and Z_k (possibly none) and updates a latent speaker, direction and turn state S_k , given the audio $X_{\leq k}$ and its committed output $H_k = (Y_{< k}, Z_{< k})$: its policy π models $P(Y_k, Z_k, S_k \mid X_{\leq k}, H_k)$. We compare policies by quality under a lag budget, $\max_{\pi} Q(\pi)$ s.t. $\text{StreamLAAL}(\pi) \leq \Lambda$ (Appendix D), and report EndOffset, switch latency and wrong-direction words alongside. A cascade factorises π into a segmenter and router, which fix a hard decision $\hat{S}_k$ without reading the translation $Y_{< k}$, and a translator, which commits under $P(Y_k \mid \hat{S}_k, X_{\leq k}, H_k)$ but cannot correct $\hat{S}_k$. We hypothesise that the two decisions are better made jointly: lexical completion helps to predict turn ends (Ekstedt & Skantze,

Table 2. CST-Bench composition (TAXI: CST-Bench-TAXI; Syn: CST-Bench-Syn). TAXI speech, content and turn order are real; its timing is synthetic. The TAXI conditions render the same turns, so merged cells hold for all three. Pairs are De→En/En→De (German/English source turns). Speech is counted after silence trimming. Turns/session: median (mean; range); switches: total (median per session). Gaps are taken at speaker changes; occupancy is the share of session time covered by turns. *Planned; blank cells are pending. †English side human, Korean side machine-translated and consensus-checked.

	TAXI (real speech)			Syn (TTS)*	
	L0 natural	L0 mediated	L1 turn-end overlap*	En↔De	Ko↔En
Source speech	telephone, 8→16 kHz			TTS	TTS
References	human			human	mixed†
Sessions	86				
Usable turns	640 (349 / 291)				
Speech (min)	40.378 (17.953 / 22.425)				
Source words	6,815 (3,228 / 3,587)				
Reference words	6,617 (3,339 / 3,278)				
Mean turn (s)	3.086 / 4.624				
Turns/session	7 (7.44; 1–17)				
Switches	540 (6)				
Session audio (h)	0.737	0.998			
Median session (s)	29.023	39.316			
Gap mean (s)	0.257	1.995			
Gap median (s)	0.223	2.026			
Overlap ratio	0.0	0.0			
Occupancy (%)	91.27	67.42			

2020), and a turn end forces the rest of its translation to be committed. Section 5 describes a model that makes both decisions from one state, and Section 6.6 compares it with cascades and ablations.

4. CST-Bench

4.1. Design principles and release

CST-Bench renders two-language dialogues as continuous single-channel sessions, varying only the turn-taking, and scores what a system commits, and when (Figure 2 and Table 2). It serves evaluation only: a calibration split of about 5% of the synthetic sessions sets wrapper thresholds and latency settings by one rule for all systems, and nothing is tuned on TAXI. [Draft: the calibration split is planned with CST-Bench-Syn] Any system that emits committed, time-stamped pieces can be scored, e.g., through simulstream (Gaido et al., 2025). Code (Apache-2.0), frozen configurations and checksums define the release, with timeline digests as the reproducibility contract; TAXI audio, free under a scientific licence but not redistributable, is rebuilt from its original distribution.

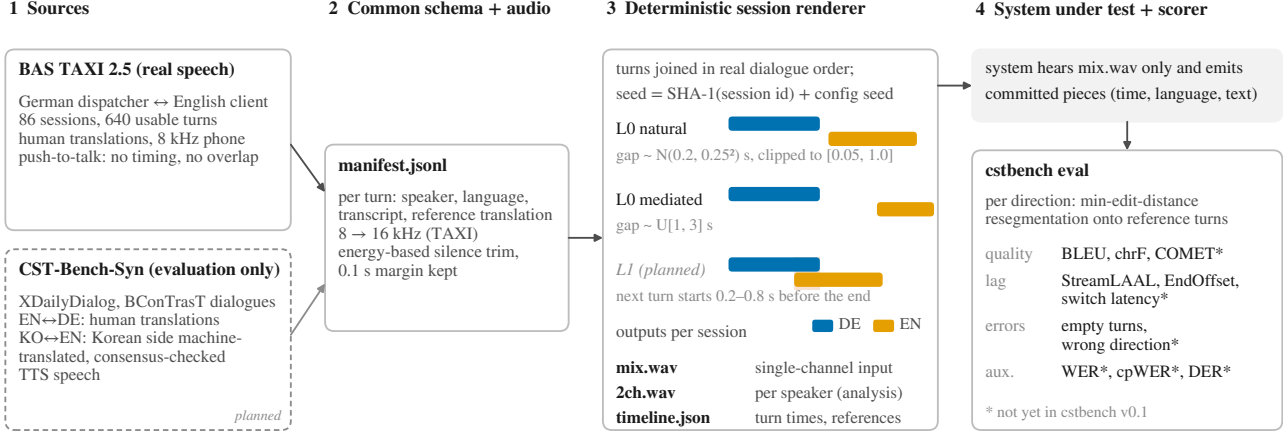
4.2. Real track: BAS TAXI 2.5

In CST-Bench-TAXI, speech, content, turn order and translations are real, but the timing is synthetic. TAXI 2.5 (Bavarian Archive for Speech Signals (BAS), 2001; Reichel et al., 2016) holds telephone dialogues between German-speaking taxi dispatchers and English-speaking clients, with human translations of all usable turns (those with a non-empty cleaned transcript and a translation): 640 of 697 (91.82%) in 86 sessions. Push-to-talk recording left one file per turn, without timing or overlap. Resampled from 8 to 16 kHz (telephone band kept) and silence-trimmed, the turns hold 40.378 min of speech (52.631 min of usable turn audio before trimming). Their mono mix simulates a shared device rather than the original calls, but lacks far-field room acoustics.

Turns and sessions are short (median 2.86 s, 22.5% above 5 s; median 7 turns and 6 direction switches per session). Language coincides with role, and the German dispatcher opens 85 of 86 sessions; as language identification also reveals a turn’s direction, we include language-ID routing baselines and stress timing over direction detection. Two single-turn sessions are excluded from switch metrics, and about 3.3k reference words per direction call for session-clustered confidence intervals. A local rebuild matched all 86 timeline digests and audio hashes of both L0 configurations. No TAXI text appears in this paper (datasheet in Appendix A).

4.3. Synthetic track: CST-Bench-Syn

[Draft: planned: nothing in this track is built yet] CST-Bench-Syn is evaluation-only synthetic speech and cannot replace a real test set, but it adds overlap measurable at the word level, both speaker–language assignments (removing TAXI’s first-speaker prior) and a distant pair, Ko↔En; its En↔De rankings are checked against CST-Bench-TAXI (Appendix C). Speakers A and B take their turns from the two language versions of one dialogue in XDailyDialog (Liu et al., 2023; Li et al., 2017), translated professionally, or BConTrasT (Farajian et al., 2020), machine-translated into German and post-edited by native speakers, so each version supplies the other’s references. Korean text is machine-translated from English, kept where two of three systems agree under quality estimation (Juraska et al., 2024; Rei et al., 2022b), and released from a model family that no evaluated system uses. As one side of every pair is translationese, we never compare absolute scores across directions. A commercial TTS edition (Microsoft, 2026), whose word boundaries are checked against forced alignment, carries the main results and an open one (Du et al., 2025; Hu et al., 2026) tests rank agreement, without cloning real voices (Appendix B).



Released: code (Apache-2.0), configs and checksums only; TAXI audio and text are never redistributed (rebuild verified: 86/86 timeline digests).

Figure 2. Building and scoring CST-Bench. Real TAXI turns and same-context synthetic dialogues (planned; dashed) are converted to one schema and rendered deterministically into continuous sessions under a timing condition: L0 natural (speaker-change gap $\sim \mathcal{N}(0.2, 0.25^2)$ s clipped to [0.05, 1.0] s), L0 mediated ($\mathcal{U}[1, 3]$ s) or L1 turn-end overlap (the next turn starts 0.2–0.8 s before the previous one ends; planned). Systems hear only `mix.wav` and emit committed, time-stamped pieces whose language encodes the direction; `cstbench eval` resegments them per direction onto the reference turns. For TAXI, only code, configurations, checksums and per-turn scores keyed by turn ID are released. *Not yet implemented in `cstbench v0.1`.

4.4. Timing conditions and deterministic rendering

As for simulated diarization data (Landini et al., 2022), sessions are assembled from turn audio with controlled timing, here deterministically (seeds derived from session identifiers; Appendix A). Energy-trimmed turns, whose intervals keep 0.1 s margins, are placed in dialogue order after a 0.5 s lead, with $\mathcal{U}[0.3, 0.8]$ s pauses within a speaker; as pauses and gaps separate these intervals, silences between energetic frames are up to 0.2 s longer than drawn. *L0 natural* draws speaker-change gaps from $\mathcal{N}(0.2, 0.25^2)$ s clipped to [0.05, 1.0] s, after the gaps of about 200 ms found across languages (Levinson & Torreira, 2015; Stivers et al., 2009); on CST-Bench-TAXI, 23.89% of them sit at the floor. *L0 mediated* draws them from $\mathcal{U}[1, 3]$ s. Neither overlaps (ratio 0.0 in all 86 sessions), unlike real conversation (Heldner & Edlund, 2010), so *L1* starts each new speaker’s turn $d \sim \mathcal{U}[0.2, 0.8]$ s before the previous one ends and sums the signals: real speech with synthetic overlap on CST-Bench-TAXI. Oracle segments cut from an L1 mix then contain the other speaker, and cuts from the per-speaker channels give an oracle-separation diagnostic. Overlap is reported per turn and per word, from forced alignment (Shi et al., 2026) or TTS word boundaries. *[Draft: planned: L1, with a cap for short turns fixed before any baseline run, and word-level overlap; to decide: whether d is measured between energetic frames or between margin-inclusive intervals, which leaves up to 0.2 s less overlapping speech]*

4.5. Scoring protocol and its validation

Interface and resegmentation. Submissions list committed pieces as (session, output language, emission time, text) on the ideal or a computation-aware clock. Per session and output language, `cstbench eval` resegments the hypothesis onto the reference translations of that direction by word-level Levenshtein alignment, as `mwerSegmenter` does (Matusov et al., 2005) in stream-level evaluation (Iranzo-Sánchez et al., 2021; Papi et al., 2024); words keep their piece’s emission time, and inserted words join the turn of the preceding word.

Quality. Per direction, we report COMET (wmt22-comet-da; Rei et al., 2020; 2022a), chrF (Popović, 2015) and BLEU with sacreBLEU signatures (Papineni et al., 2002; Post, 2018) on hypotheses normalised to the TAXI reference style (lowercase, unpunctuated, numbers spelled out). COMET is primary on TAXI only if it ranks systems as chrF does; reference-free CometKiwi (Rei et al., 2022b) checks the machine-translated Korean references.

Latency. For turn n of a direction, with resegmented hypothesis words $j = 1, \dots, |h_n|$ emitted at $\tau_{n,j}$, let $d_{n,j} = \max(0, \tau_{n,j} - b_n)$, $L_n = e_n - b_n$ and $\rho_n = L_n / \max(|h_n|, |r_n|)$ (lengths in words). Then

$$\text{StreamLAAL}_n = \frac{1}{J_n} \sum_{j=1}^{J_n} (d_{n,j} - (j-1)\rho_n), \quad (1)$$

$$\text{EndOffset}_n = \tau_{n,|h_n|} - e_n, \quad (2)$$

$$\text{SwitchLatency}_n = d_{n,1} \quad \text{for } n > 1, s_n \neq s_{n-1}, \quad (3)$$

where J_n is the first j with $d_{n,j} \geq L_n$ (else $|h_n|$). Equation (1) is LAAL (Papi et al., 2022), a length-adaptive Average Lagging (Ma et al., 2019), on resegmented turns, i.e., StreamLAAL (Papi et al., 2024): lag *within* a turn. Turn-end output scores the turn duration, so offline systems on oracle turns sit at the mean turn length (4.624 s for En→De, 3.086 s for De→En). StreamLAAL ignores the words after J_n , so it misses how long a translation runs on after the speaker stops; EndOffset measures this wait per turn, as SimulEval’s EndOffset (Ma et al., 2020a) and the ending offset reported for SeamlessStreaming (Seamless Communication et al., 2023) do per segment, and switch latency is SimulEval’s StartOffset at direction changes. We report means, medians and 90th percentiles over non-empty turns, always with the empty-turn rate (Appendix C).

Errors and auxiliary outputs. A committed piece is *wrong-direction*, with all its words, if turns are open at its emission time but none is translated into its language, if a text language identifier assigns it to the other language, or if it copies an open turn’s transcript in its language (Appendix C). We report such words as a share of committed words, checked against gold turn IDs where available, with empty turns and the length ratio. Auxiliary outputs are scored by WER per language, cpWER (Watanabe et al., 2020), tcpWER (von Neumann et al., 2023) and DER (columns 0 and 0.25 s; 80 ms frames).

Statistics and validation. Confidence intervals (95%) and paired tests use a session-clustered paired bootstrap (Koehn, 2004) with 2,000 resamples and Holm correction over main comparisons fixed in advance. Controlled perturbations of the reference output test whether each metric moves only as expected (Table 10). Resegmented latency has a segmentation-related bias (Polák et al., 2025) and can reward degenerate output: in a preliminary run, a copied source scored a lower De→En StreamLAAL than the reference emitted at turn ends (2.276[†] vs. 3.086[†] s; [†]preliminary, to be re-run under the final protocol), so latency is always shown with quality, length ratio and empty turns, and also on exactly matched words. We keep StreamLAAL, the only latency metric of the IWSLT 2025 simultaneous track (Abdulmumin et al., 2025), as it shares the resegmentation of our other metrics; Appendix C compares it with LongYAAL (Polák et al., 2025), primary at IWSLT 2026 (Adelani et al., 2026). *[Draft: scorer v0.1 computes BLEU, chrF, StreamLAAL, EndOffset, empty turns and wrong-language words given gold turn ids; the other metrics, the bootstrap and a pinned number normaliser are being added; rows 3–10 of Table 10 are planned]*

5. MODEL: Conversational Delayed Streams

We cast CST as serialized multi-channel emission on one clock (Figure 3): after each 80 ms audio embedding, a causal speech LLM writes zero or more tokens for a speaker-attributed transcript and for a translation into each language. Each channel trails its evidence by its own delay, so the read/write policy becomes next-token prediction against targets that, relative to a teacher’s alignment, do not anticipate unheard speech. These *conversational delayed streams* extend delayed streams modeling (Zeghidour et al., 2025), the per-word delayed translation targets of Hibiki (Labiausse et al., 2025) and serialized output training (Kanda et al., 2022a; Papi et al., 2023b) to channels with their own delays and commit points. *[Draft: planned: only the backbone of Section 5.1 is built; Sections 5.2 to 5.4 describe the design, and no CST result exists yet]*

5.1. Streaming backbone

The encoder, Nemotron 3.5 ASR Streaming 0.6B (NVIDIA, 2026), is a cache-aware (Noroozi et al., 2024) FastConformer (Gulati et al., 2020; Rekesh et al., 2023) with 24 layers, $d_{\text{model}} = 1024$ and 80 ms frames; its attention context $[56, 0]$ spans 4.48 s to the left and none to the right, and a truncation test bounds its effective look-ahead by 80 ms. An MLP adapter (LayerNorm, Linear, GELU, Linear) maps frame k to an embedding a_k that replaces an audio placeholder token of the Qwen3-ASR language model, the *thinker* (Shi et al., 2026; Xu et al., 2025); we drop Qwen3-ASR’s own non-causal audio encoder and use the 1.7B thinker (a 0.6B one also exists). After a prefix with the language and a delay token $\langle \text{DELAY}_\delta \rangle$, each chunk holds a_k , the text tokens due in it, and $\langle \text{next} \rangle$ (“read 80 ms more”), the analogue of a transducer blank (Graves, 2012; Seide et al., 2024). A word that ends at time t (by forced alignment) is due, with all its BPE tokens, in chunk

$$k(t; \delta) = \lfloor t/\Delta \rfloor + \delta, \quad \Delta = 80 \text{ ms}, \quad (4)$$

with $\delta \in \{2, 3, 4, 6, 8\}$ sampled per sequence (Figure 11). The labels, at positions $\mathcal{Y}$ of the token sequence o , are the text tokens and $\langle \text{next} \rangle$; as $\langle \text{next} \rangle$ makes up 70–85% of them, the loss is a weighted cross-entropy,

$$\mathcal{L} = - \frac{\sum_{m \in \mathcal{Y}} w_m \log p_\theta(o_m | o_{<m})}{\sum_{m \in \mathcal{Y}} w_m}, \quad (5)$$

with $w_m = 1$ for text and $w_m = 0.3$ (English) or 0.15 (Korean) for $\langle \text{next} \rangle$; a final (non-streaming) mode is trained jointly.

In earlier backbones with the 0.6B thinker, the delay token set the median emission delay $(k+1)\Delta - t$ (without encoder look-ahead) to about 200/280/360 ms for $\delta = 2/3/4$; after one epoch, the 1.7B backbone reaches an

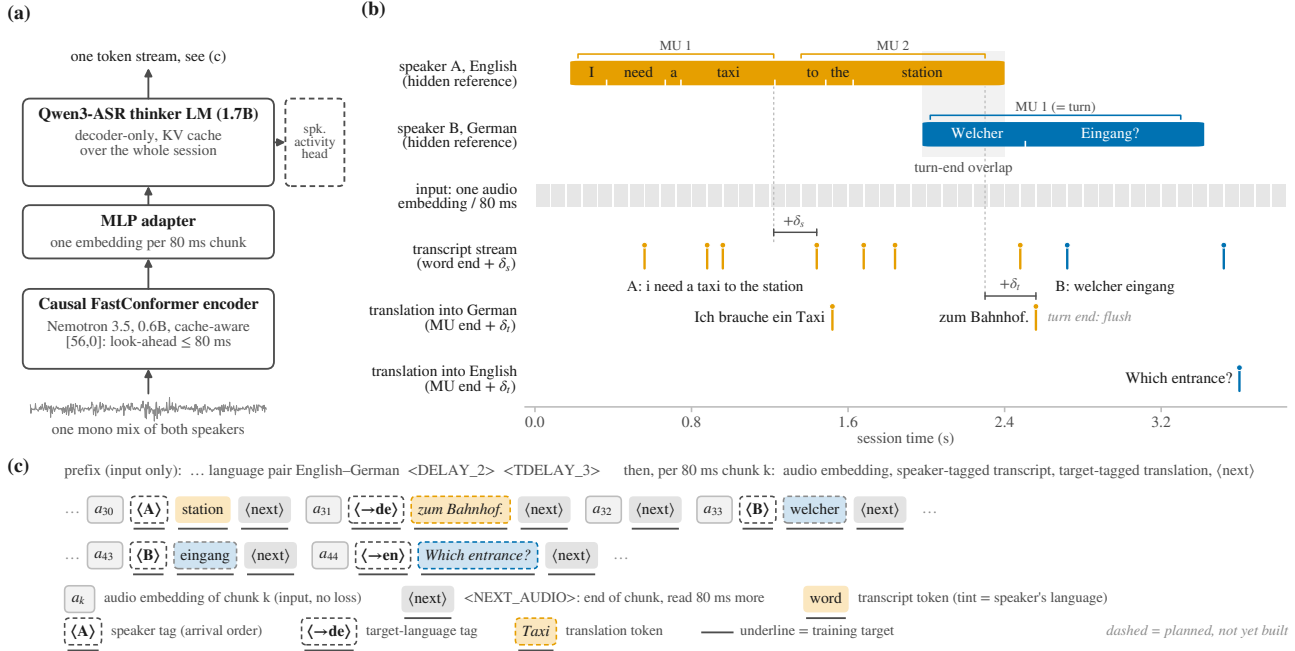


Figure 3. MODEL and conversational delayed streams (illustrative timings; $\delta_s = 2$ and $\delta_t = 3$ chunks of 80 ms). (a) A causal, cache-aware FastConformer encoder feeds one adapter embedding per 80 ms chunk to a Qwen3-ASR thinker LM that keeps the whole session in its KV cache; a two-slot head gives frame-level speaker activity for DER. (b) Targets on the 80 ms clock: a transcript word is emitted δ_s chunks after the chunk in which it ends, the translation of each verified meaning unit (numbered 1, 2, ... in the figure) δ_t chunks after the chunk in which the unit ends, and a turn end forces a commit. (c) The resulting token sequence: a_k , audio embedding of chunk k (input, no loss); $\langle A \rangle$, $\langle B \rangle$, arrival-order speaker tags; $\langle \rightarrow de \rangle$, $\langle \rightarrow en \rangle$, target-language tags; $\langle next \rangle$, end of chunk; underlined tokens are training targets. Dashed parts are planned and not yet built. [Draft: built today: the encoder–adapter–LM backbone, trained for single-speaker English and Korean transcription]

English VoxPopuli-AA WER of 7.70/7.18/6.84% at $\delta = 4/\delta = 6/\text{final}$, against 6.88% for the same encoder with its causal RNN-T decoder and 5.10% for the non-causal offline Qwen3-ASR-1.7B (Table 17). A [56, 3] right context and a larger δ are largely interchangeable at matched mean delay (Figure 12), so the encoder stays causal. All of this is single-speaker English and Korean transcription (Appendix F).

5.2. One serialized stream

MODEL keeps this clock (Figure 3c); its prefix names the language pair and two delays, $\langle \text{DELAY}_{\delta_s} \rangle$ for transcripts and $\langle \text{TDELAY}_{\delta_t} \rangle$ for translations, sampled per training session. Chunk k reads

$$B_k = a_k \langle \hat{s}_1 \rangle z_1 \langle \hat{s}_2 \rangle z_2 \cdots \langle \rightarrow \lambda_1 \rangle y_1 \langle \rightarrow \lambda_2 \rangle y_2 \cdots \langle next \rangle, \quad (6)$$

where z_j are words of speaker $\hat{s}_j \in \{A, B\}$, numbered by arrival without enrolment, and y_i translates one unit into language λ_i , which fixes the direction; either part may be empty. Each tagged segment is one committed piece (Section 3), so the latent speaker and direction state is explicit. Under turn-end overlap, the words of both speakers are ordered by end time, as in t-SOT (Kanda et al., 2022a) and

in JSTAR’s speaker-labelled targets (Moritz et al., 2025). Tags count as text in Equation (5); a two-slot activity head on the audio positions adds a binary cross-entropy and serves DER, with arrival-order slots (Park et al., 2025) instead of permutation-free training (Fujita et al., 2019). The KV cache holds the whole session, so both sides of the dialogue condition every decision. The longest TAXI session (90.02 s) spans about 1.1k audio positions. [Draft: planned: the two speaker tags are reserved in the vocabulary but untrained; translation segments, $\langle \text{TDELAY}_{\delta_t} \rangle$ and the activity head do not exist yet]

5.3. Commit targets

Transcript words are force-aligned per speaker before mixing (Qwen3-ForcedAligner; Shi et al., 2026) and placed by Equation (4) with $\delta = \delta_s$. For translations, a teacher LLM (Qwen3.8-27B) translates the source turn $u_{1:I}$ into v , and contextual alignment (Labiausse et al., 2025) maps target word v_j to $\alpha_j = \arg \max_i [\phi_j(i) - \phi_j(i-1)]$, where $\phi_j(i)$ is the log-probability of v_j when v is force-decoded from $u_{1:i}$. Source position i is a candidate if the words with $\alpha_j \leq i$ form a prefix of v longer than the committed one (the alignment analogue of meaning units; Zhang et al., 2020) and if the source has grown by a minimum

unit since the last commit (pilot: 2 Korean or 3 English words, and 1.0 s). The teacher then translates only the new part, given the source so far and the committed text; the unit is accepted if this text is in the target language and reaches a chrF of at least 50 against the newly aligned span of v . We call accepted units verified meaning units (MU2). A turn end flushes the rest of the turn, so the model must learn to recognise turn ends. A unit whose source ends at time e is written in chunk $\lfloor e/\Delta \rfloor + \delta_t$, with $\delta_t \geq \delta_s$ so that its transcript comes first. Relative to the teacher’s alignment, no target thus anticipates unheard speech, as with Hibiki’s per-word delays, whereas sentence-level targets used with prefix-to-prefix training (Ma et al., 2019) demand anticipating German clause-final verbs or Korean predicates. The commit points, however, are chosen with the whole turn in view, so MODEL imitates a privileged teacher, which Table 3 scores on gold transcripts as Gold→LLM (MU2). In a text-level Ko↔En pilot with silver references (Appendix G), MU2 roughly halves computation-unaware LAAL (Papi et al., 2022), here normalised by the output length, relative to commits at the utterance end (11.09→5.19 s Ko→En; 15.05→7.33 s En→Ko), while chrF falls from 61.3 to 60.2 and from 44.2 to 42.0. Utterance-end LAAL equals the mean utterance length, whereas CST-Bench-TAXI turns average 3.086 s (German) and 4.624 s (English), which bounds what commits inside a turn can gain there (Figure 10). Unverified alignment-only units, continued from a forced prefix, collapsed in 31 of 150 En→Ko outputs, MU2 in one. [Draft: planned: MU2 targets for En↔De and at training scale; the German minimum unit and the δ_t values are set on held-out data]

5.4. Training without real cross-lingual conversations

As TAXI is test-only and we know of no two-language conversations with overlap and translations at training scale, MODEL learns from single-speaker corpora and simulated sessions in stages (Table 16). After an adapter-only warm-up (Stage 0), Stage 1 trained the backbone on about 3.8k h + 4.3k h of English and Korean speech. Stage 2 trains M1, which transcribes and attributes speakers: it adds German ASR data, 8 kHz telephone-band augmentation, speaker tags and the activity head on simulated two-speaker En↔De and Ko↔En mono sessions with gaps and overlaps drawn more broadly than in the benchmark (cf. Landini et al., 2022). Stage 3 trains M2 and M3, which add translation streams committed at turn ends (M2) or at MU2 (M3), from human En↔De speech translation data (Wang et al., 2021; Di Gangi et al., 2019) and silver teacher translations. Unlike the off-the-shelf baselines, MODEL thus sees benchmark-like simulated sessions, as does the same-backbone cascade through M1 (Section 6.6). The fixed-gap sweep (Figure 9) tests whether MODEL’s gains depend on the benchmark’s gaps. Training excludes TAXI, the dia-

logue corpora and voices of CST-Bench-Syn, and outputs of EXAONE-4.0, an LLM we use only for evaluation (Appendices B and G). [Draft: planned: Stages 2–3; German training data has yet to be obtained]

5.5. Inference and latency accounting

Decoding is greedy, with one forward pass per chunk plus one per emitted token. A piece written after chunk k is time-stamped $(k+1)\Delta + \ell_{\text{enc}} + c$, so a token emitted on schedule after evidence that ends at t has latency $q + \delta\Delta + \ell_{\text{enc}} + c$: frame quantization $q = (\lfloor t/\Delta \rfloor + 1)\Delta - t \in (0, \Delta]$, stream delay $\delta \in \{\delta_s, \delta_t\}$, encoder look-ahead ℓ_{enc} (counted at its 80 ms bound) and compute time c , which the ideal clock sets to zero. As a turn end forces a commit, a model that follows its targets keeps EndOffset near $q + \delta_t\Delta + \ell_{\text{enc}} + c$. Real time is not yet shown on GPUs: on an H200, the 0.6B backbone needs 100–150 ms per 80 ms chunk at the 99th percentile, and the 1.7B thinker is unmeasured (Table 14). Translation segments need a higher cap than the backbone’s 8 tokens per chunk. [Draft: planned] The encoder is licensed under OpenMDW-1.1 and the thinker under Apache 2.0.

6. Experiments

We answer RQ1–RQ4 (Section 1) on real dialogues first; synthetic sessions add the distant pair and a check against real rankings. Every comparison carries a session-clustered 95% confidence interval (CI; Appendix H), and we do not interpret differences inside it. [Draft: planned: apart from the preliminary offline rows of Table 3 and ASR WERs of Table 5, no experiment of this section has been run]

6.1. Setup

Systems. Every system receives the session’s language pair. Offline systems cut each oracle turn from the mix and translate it after the turn ends, one turn per prompt and without dialogue history: Qwen3.8-27B (“LLM”) translates the gold transcript (Gold→LLM) or the transcript of Qwen3-ASR-1.7B, given the language (ASR→LLM; Shi et al., 2026), and Whisper-large-v3 translates speech into English (Whisper ST; Radford et al., 2023). Gold→LLM (MU2) is a streaming variant with oracle input: the LLM reads gold transcripts at forced-aligned word times and commits at the verified meaning units (MU2) of Section 5, whose boundaries are found with the whole turn in view (non-causal). The consecutive cascade detects turn ends with the realistic wrapper described below, then transcribes and translates each segment like ASR→LLM. The streaming systems are SeamlessStreaming (Seamless Communication et al., 2023), StreamSpeech (Zhang et al., 2024) (De→En only), InfiniSST (Ouyang et al., 2025) (En→De only) and two systems that run offline translation models with a streaming policy. In the streaming cascade, the LLM

Table 3. Real dialogues: CST-Bench-TAXI, L0 natural timing, ideal clock. Rows: upper bounds with oracle input, the consecutive cascade, streaming systems and MODEL (Section 6.1; LLM: Qwen3.8-27B), each at its main operating point; MODEL M2 commits translations at turn ends, and M3 at MU2 units with a low or a high δ_t . Seg.: oracle = gold turns and source language; realistic = VAD and language-ID routing; none = the system segments the stream itself. Per direction: mean StreamLAAL and median EndOffset in seconds; the last three columns pool both directions (switch latency: median, in seconds) and do not apply to single-direction systems, which cannot be the best pipeline (Table 4 and Figure 5). Table 13 gives 95% session-bootstrap CIs for the rows it repeats, except for BLEU. Blank: pending; -: direction not covered. [†]preliminary and scored by gold turn ID (oracle mode of Appendix C), not resegmented; to be re-run under the final protocol. [‡]0 by construction: the output is emitted at the oracle turn end.

System	Seg.	En→De					De→En					Both directions		
		COMET	BLEU	chrF	Stream-LAAL	End-Offset	COMET	BLEU	chrF	Stream-LAAL	End-Offset	Wrong dir. (%)	Switch latency	Empty turns (%)
Gold→LLM	oracle		48.31 [†]	71.72 [†]	4.62 [†]	0 [‡]		30.60 [†]	58.05 [†]	3.09 [†]	0 [‡]			
ASR→LLM	oracle		42.66 [†]	67.97 [†]	4.62 [†]	0 [‡]		28.11 [†]	56.61 [†]	3.09 [†]	0 [‡]			
Whisper ST	oracle	–	–	–	–	–		30.69 [†]	56.67 [†]	3.09 [†]	0 [‡]	–	–	–
Gold→LLM (MU2)	oracle													
Consecutive cascade	realistic													
SeamlessStreaming	oracle													
	realistic													
2×SeamlessStreaming	none													
Streaming cascade	oracle													
	realistic													
M4T v2 + AlignAtt	oracle													
	realistic													
StreamSpeech	oracle	–	–	–	–	–						–	–	–
	realistic	–	–	–	–	–						–	–	–
InfiniSST	oracle						–	–	–	–	–	–	–	–
	realistic						–	–	–	–	–	–	–	–
MODEL M2	none													
MODEL M3, low δ_t	none													
MODEL M3, high δ_t	none													

re-translates the growing transcript of the Nemotron 3.5 streaming RNN-T (NVIDIA, 2026) and commits the prefix on which two consecutive outputs agree (LocalAgreement-2, LA-2; Liu et al., 2020; Polák et al., 2022; Macháček et al., 2023). M4T v2 + AlignAtt runs SeamlessM4T v2 (SEAMLESS Communication Team et al., 2025; Seamless Communication et al., 2023) with the AlignAtt policy (Papi et al., 2023a) in simulstream (Gaido et al., 2025). As a naive bidirectional baseline, 2×SeamlessStreaming runs one instance per target language on the whole mix, without routing. MODEL runs as M2, which commits translations at turn ends, and as M3, which commits at MU2 units and runs at a low and a high δ_t (Section 5). Table 12 lists checkpoints, sizes and licences. [Draft: *InfiniSST only if its weights are released; Canary-IB-v2 (Sekoyan et al., 2025) with AlignAtt is optional; Hibiki-Zero (De→En; public weights) is not yet in the plan*]

Wrappers and operating points. A system that translates one direction runs as one instance per target language behind a wrapper; we call the combination a pipeline (Appendix D). The oracle wrapper cuts gold turns from the mix and routes each by its gold source language. The realistic wrapper segments the stream with Silero VAD (Silero Team, 2024) and routes each segment by spoken

language identification restricted to the pair, decided at the segment start and re-routed if the decision flips. Baselines are not fine-tuned. Each streaming system has two operating points, one of them its main point, chosen once on the CST-Bench-Syn calibration split by a rule fixed before any test run. [Draft: *until CST-Bench-Syn exists, documented defaults stand in for calibrated operating points; which of the two latency budgets gives the main point is still to be fixed*]

Metrics and statistics. Per direction, we report COMET, chrF and BLEU, StreamLAAL (lag within a turn) and median EndOffset (the listener’s wait after a turn); pooled over directions, we report wrong-direction words, median switch latency and empty turns (Section 4). We rank systems by COMET rather than BLEU (Kocmi et al., 2021); on CST-Bench-TAXI, whose references are lowercase and unpunctuated, chrF takes its place if the two rank systems differently (Section 6.7). Latency uses the ideal clock; computation-aware latency, measured for one unbatched stream on one GPU type, is in Table 14. The oracle-gap recovery of MODEL is $R = (Q_{\text{ours}} - Q_{\text{real}})/(Q_{\text{oracle}} - Q_{\text{real}})$, where Q is COMET: of MODEL, and of the pipeline of Table 4 under the oracle and the realistic wrapper; R comes with a bootstrap CI and is reported only where the CI of the gap excludes 0. Daggers mark preliminary num-

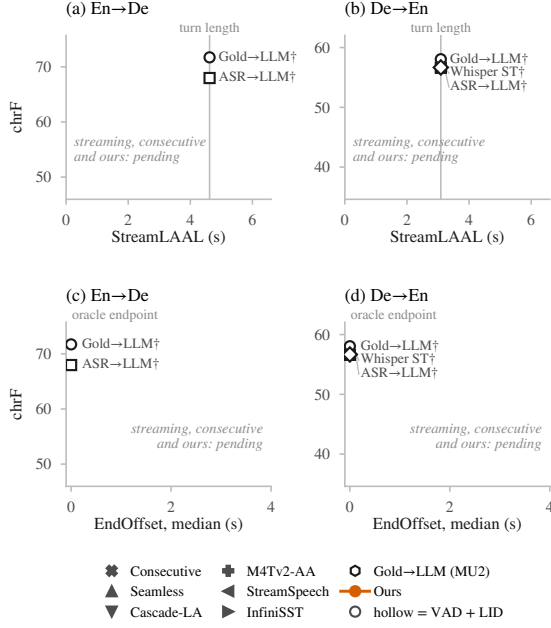


Figure 4. Quality vs. latency on CST-Bench-TAXI (L0 natural, ideal clock), per direction: chrF against StreamLAAL (a, b) and against median EndOffset (c, d). Black outlines mark systems with oracle input, although drawn hollow; by construction, the labelled offline systems sit at the mean source-turn duration on StreamLAAL (‘turn length’) and at 0 on EndOffset (‘oracle endpoint’). The other systems are pending: filled markers will show oracle segmentation and hollow markers VAD and language-ID routing, with grey arrows joining the two for each system, and a vermilion curve will trace MODEL over its δ_t operating points. Legend: Seamless = SeamlessStreaming, Cascade-LA = streaming cascade, M4Tv2-AA = M4T v2 + AlignAtt, Ours = MODEL. chrF stands in until COMET is in the scorer. [†]preliminary; to be re-run under the final protocol.

bers, to be re-run under the final protocol.

6.2. Real dialogues (RQ1, RQ3)

Table 3 and Figure 4 compare all systems on CST-Bench-TAXI under L0 natural timing. Given oracle turns, Gold→LLM reaches 48.31[†] BLEU and 71.72[†] chrF for En→De, and 30.60[†] and 58.05[†] for De→En. ASR transcripts cost 5.65[†] and 2.49[†] BLEU, in line with the higher WER of Qwen3-ASR-1.7B on English than on German turns (13.78% vs. 7.33%; Table 5), and Whisper ST reaches 30.69[†] BLEU into English. These offline systems look instantaneous on EndOffset, at most about 0.2[†] s even with measured, batched computation (Table 14), only because they receive oracle endpoints. The consecutive cascade, which must detect endpoints, makes listeners wait a median of [TBD] s at [TBD] COMET. Realistic segmentation costs the streaming systems [TBD] COMET (arrows in Figure 4), and MODEL reaches [TBD] COMET at a median EndOffset of [TBD] s without oracle input.

Table 4. Where does the conversational loss come from? Oracle-component factorial for the best pipeline of Table 3 (by realistic-wrapper COMET on CST-Bench-TAXI L0 natural). Oracle components (✓): T, gold turn segmentation (else VAD); L, gold source language for routing (else spoken language ID; with VAD segments, the language of the turn with the largest overlap); S, two-channel separation (L1 only; else segments are cut from the mono mix). Row 4 is the realistic pipeline; row 6 is MODEL, without oracle input. COMET is the mean of the two directions; switch latency is the median in seconds; Δ COMET is COMET under L1 minus that of row 1. Blank: pending; -: not applicable.

	Oracle	L0 natural		L1 turn-end overlap			
		T	L	S	Wrong dir. (%)	Wrong Switch latency	Δ COMET
#							
1	✓	✓	✓	✓			–
2	✓	✓	✓	✓			
3	✓	✓	✓	✓			
4	✓	✓	✓	✓			
5	✓	✓	✓	✓	–	–	
6	MODEL						

6.3. Where pipelines fail (RQ1)

Table 4 replaces one oracle component at a time for the best pipeline of Table 3; Appendix E repeats this for the second best. We expect silence-based segmentation to merge turns across speaker changes: a 500 ms threshold captures only 18–30% of between-speaker intervals in conversational corpora (Heldner & Edlund, 2010), and the median speaker-change gap of CST-Bench-TAXI L0 natural is 0.223 s. Under L0 natural, turn segmentation costs [TBD] COMET and routing [TBD], at [TBD]% language-ID routing accuracy; under L1, oracle separation recovers [TBD] COMET. Short turns are a further risk, as 59 CST-Bench-TAXI turns have at most three words; Appendix E breaks errors down by type (Table 15) and listener wait by source-turn duration (Figure 10).

6.4. Timing conditions and language pairs (RQ2, RQ3)

Figure 5 contrasts, for each timing condition and language pair, the best pipeline under both wrappers with MODEL. On CST-Bench-TAXI, the oracle-realistic COMET gap is [TBD] under L0 mediated, [TBD] under L0 natural and [TBD] under L1, and MODEL recovers [TBD]% of it under L1. Re-rendering CST-Bench-TAXI with one fixed gap from −0.6 to 2 s at every speaker change traces the dose-response (Figure 9). We compare language pairs only through each pair’s gap to its own oracle, never through absolute scores. The text-level unit pilot already shows a direction asymmetry in the distant pair: only 21% of candidate commit points pass verification for En→Ko, against 54% for Ko→En (Appendix G).

placeholder: values pending

● best oracle-segmented pipeline ○ best realistic pipeline (VAD + LID) ◆ ours (no oracle input)

TAXI (real speech)			
L0 mediated	pending	pending	pending
L0 natural	pending	pending	pending
L1 turn-end overlap	pending	pending	pending
Syn EN↔DE (TTS)			
L0 natural	pending	pending	pending
L1 turn-end overlap	pending	pending	pending
Syn KO↔EN (TTS)			
L0 natural	pending	pending	pending
L1 turn-end overlap	pending	pending	pending
	COMET (mean of the two directions)	EndOffset, median (s)	wrong-direction words (%)

Figure 5. Where the conversational difficulty lies (values pending; numbers will be in Table 13). For each timing condition and language pair: the best pipeline with oracle turn segmentation (filled) and with VAD and language-ID routing (hollow), and MODEL (diamond; no oracle input). One pipeline serves every row and both wrappers: that of Table 4, with the highest realistic-wrapper COMET on CST-Bench-TAXI L0 natural among pipelines that cover both directions, at its main operating point. Panels: COMET (mean of the two directions), median EndOffset and wrong-direction words.

Table 5. Who said what: transcription and two-speaker attribution. Oracle ASR: Qwen3-ASR-1.7B on oracle turns with the language given; offline cascade: pyannote diarization, then Qwen3-ASR-1.7B; streaming cascade: Streaming Sortformer, then the Nemotron 3.5 streaming RNN-T. Error rates in % on CST-Bench-TAXI under L0 natural (L0) and L1, and cpCER on CST-Bench-Syn Ko↔En under L1; DER with a 0.25 s collar (collar 0 in Appendix E). Delay: median emission delay of transcript words; earlier 0.6B versions of the backbone have about 200 ms at $\delta = 2$ and 360 ms at $\delta = 4$ on single-speaker English and Korean, excluding the 80 ms encoder look-ahead counted in Section 5. Blank: pending; -: not applicable (oracle speakers or offline output). [†]preliminary; to be re-run under the final protocol.

Set	Metric	Oracle ASR	Offline cascade	Streaming cascade	MODEL	
					$\delta_s=2$	$\delta_s=4$
TAXI L0	WER En	13.78 [†]				
	WER De	7.33 [†]				
	cpWER					
	DER	–				
	Delay (ms)	–	–			
TAXI L1	cpWER					
	DER	–				
Syn L1	cpCER					

Table 6. Ablations of MODEL, one change per row from the same M1 checkpoint. MODEL (M3) commits at MU2 units, uses speaker tags and sees L1 sessions in training; the next rows commit at turn ends instead (M2), keep only target-language tags, drop the L1 sessions, or train fixed- δ_t models instead of conditioning on δ_t . M1→LLM: same-backbone cascade (M1 transcripts, Qwen3.8-27B, LA-2); ASR→1.7B LLM: size-matched cascade (Qwen3-ASR-1.7B, an LLM of about 1.7B parameters, LA-2). COMET is given as En→De / De→En; StreamLAAL is the mean over both directions and EndOffset and switch latency are medians, all in seconds; wrong-direction words in %. All cells are pending. [◦]Optional run.

Variant	L0 natural		L1 turn-end overlap	
	Stream-COMET	End-Offset	Wrong dir.	Switch latency
MODEL (M3)				
turn-end commits				
no speaker tags				
no L1 in training [◦]				
fixed- δ_t models [◦]				
M1→LLM				
ASR→1.7B LLM				

[Draft: rows of runs that are cut will be removed, not left blank]

6.5. Who said what: transcripts and speakers

Table 5 scores transcripts and speaker attribution with WER per language, cpWER (Watanabe et al., 2020) and DER. The baselines are Qwen3-ASR-1.7B on oracle turns, an offline cascade of pyannote diarization (Bredin, 2023) and Qwen3-ASR-1.7B, and a streaming cascade of Streaming Sortformer (Park et al., 2025; Medennikov et al., 2025) and the Nemotron 3.5 streaming RNN-T. MODEL reaches

[TBD]% cpWER and [TBD]% DER on CST-Bench-TAXI L1 at $\delta_s = 2$. DER needs care here: in the first release each speaker keeps one language, so the language reveals the speaker, and DER mostly measures boundary timing under short gaps and overlap; speakers who switch languages, which we exclude, would break this equivalence.

6.6. Ablations: does the unified design help?

Table 6 changes one thing at a time from the same M1 checkpoint (Section 5): commits at turn ends (M2) instead of MU2; target-language tags without speaker tags, which tests whether an explicit speaker channel helps translation; and, optionally, no L1 sessions in training, or fixed- δ_t models instead of δ_t conditioning. Two cascades keep the factorisation of Section 3 at two scales: a same-backbone cascade translates M1 transcripts with the LLM under LA-2, and a size-matched cascade pairs Qwen3-ASR-1.7B with an LLM of about 1.7B parameters. We fixed in advance the comparison with the same-backbone cascade as the main test of the unified design: at matched StreamLAAL, MODEL differs from it by [TBD] COMET, [TBD] points of wrong-direction words and [TBD] s of switch latency. As this cascade also differs in translator size, commit policy and input, it tests the design as a whole, not the coupling of turn-taking and translation alone. *[Draft: undecided: a decoupled twin (same 1.7B thinker, data and MU2 commits, translating the segments fixed by M1’s speaker tags) would isolate the coupling]*

6.7. Validity checks (RQ4)

Appendix C tests whether the conclusions hold beyond one rendering and one metric (Table 11 and Figure 8). It reports Kendall’s τ (in its tie-corrected form, τ_b) between system rankings on CST-Bench-TAXI and CST-Bench-Syn En $\leftrightarrow$ De under L0 natural, between the two TTS editions, between gold and silver references and across render seeds, and the rank agreement of StreamLAAL, EndOffset and LongYAAL (Polák et al., 2025), and of COMET and chrF on CST-Bench-TAXI’s unpunctuated references. Synthetic and real rankings give a Kendall’s τ of [TBD]; we report it whatever its value.

7. Limitations and Conclusion

Data. CST-Bench-TAXI has real speech, content and turn order, but its push-to-talk turns leave every gap synthetic and nothing overlapping; it is also small (about 3.3k reference words per direction), single-domain and narrow-band, and ties role to language (the German dispatcher speaks first in 85 of 86 sessions). We know of no public recordings of two-language conversation with natural overlap and reference translations; the closest, InCroMin (Čechovič et al., 2025), is mediated by machine translation, recorded per participant and has no English–German pair. We therefore control timing, add overlap synthetically, report session-clustered confidence intervals, include language-ID routing, and render CST-Bench-Syn with both speaker–language assignments. CST-Bench-Syn lacks real disfluencies (the transcripts of the 640 TAXI turns mark 60 German and 172 English filled pauses), is translationese on

one side, and has consensus-filtered machine-translated Korean; we draw only relative conclusions and test whether it ranks systems as CST-Bench-TAXI does (Table 11). *[Draft: planned; CST-Bench-Syn and the turn-end overlap renders are not built yet]*

Measurement and scope. Main-text latencies use the ideal clock; real-time claims rest on Table 14, where the 0.6B backbone still misses its 80 ms budget at the 99th percentile. Energy-trimmed turn boundaries may shift latency reference points by up to about 0.1 s until forced alignment replaces them (Appendix A). Human interpreters, with an English-to-Korean ear-voice span of about 3 s (Lee, 2002), are a reference point, not a target. Interruptions and barge-in are excluded for lack of real recordings to validate them, and content-free backchannels (L1b) are at most an optional stress test. With one language per speaker, diarization reduces largely to language-aware segmentation, so DER mainly measures boundary timing (Table 5).

Unified model. MODEL learns from simulated sessions and teacher-made commit targets, which are anticipation-free only relative to the teacher’s alignment; same-backbone and size-matched cascades separate design from scale (Table 6). *[Draft: planned; only the streaming ASR backbone is built, and all MODEL results are pending]*

Conclusion. CST requires deciding, from one mono stream, who speaks, which way to translate, and when to commit. Segmentation and routing explain [TBD]% and [TBD]% of the oracle–realistic gap (RQ1), which moves from [TBD] to [TBD] COMET between mediated gaps and turn-end overlap (RQ2); MODEL recovers [TBD]% of it without oracle input, at a median EndOffset of [TBD] s (RQ3); and synthetic and real sessions give a Kendall’s τ of [TBD] between system rankings (RQ4). Next come interruptions and backchannels, real overlapped bilingual recordings, and more language pairs.

Impact Statement

CST systems could let people who share no language talk through one device, and speaker-attributed transcripts could help deaf and hard-of-hearing people follow bilingual conversations. A mistranslated or misattributed turn can mislead in medical or legal settings, so deployments there should show the source transcript next to each translation and keep human interpreters available. Transcribing and attributing speech from one microphone could also enable covert monitoring; MODEL is designed to label speakers only by order of arrival, without identifying them, and any deployment should require the consent of everyone recorded. We redistribute no TAXI audio or text (only build scripts, configurations, checksums and per-turn scores),

and all text examples in this paper are invented, except for one KsponSpeech transcript in Figure 11.*[Draft: check that the KsponSpeech terms allow quoting this transcript; otherwise use an invented utterance there]* CST-Bench is intended for evaluation only; its synthetic voices are disclosed and clone no real person, and its XDailyDialog-based subset is non-commercial.*[Draft: planned; CST-Bench-Syn is not built yet]* Outputs of EXAONE-4.0, whose licence is non-commercial, serve only internal evaluation and are neither released nor used for training, and MODEL combines components licensed under OpenMDW-1.1 and Apache 2.0 (Table 19).

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A. TAXI Datasheet and Why TAXI

This appendix documents CST-Bench-TAXI, the real track of CST-Bench (Section 4), and explains why it is built from TAXI. Table 7 lists its statistics, Figure 6 shows their distributions and Figure 7 shows one rendered session. Because TAXI may not be redistributed, we report aggregates only: no TAXI audio, transcript or translation appears in this paper, its figures or the released files.

Table 7. Full statistics of the CST-Bench-TAXI build (cstbench v0.1.0). Rendered turns are the 16 kHz turns after silence trimming, 0.1 s margins included; they are identical in L0 natural and L0 mediated. Untrimmed values refer to the usable source turn files. Turn files exclude pre-dialogue and hang-up items. Same-speaker successions: 9 arise because unusable turns between them were skipped, 5 are adjacent in the source. Reference words are those of the human translations of the column’s turns. Marks (SpeechDat notation): * hardly recognisable, ~ truncated, ** unintelligible. Percentiles use linear interpolation; the scorer’s p90 of m values instead takes the sorted value at the 0-based index $\min(m - 1, \lfloor 0.9m \rfloor)$ (Appendix C). –: not applicable or not reported.

Statistic	German (dispatcher)	English (client)	All
<i>Sessions and turns</i>			
Sessions	–	–	86
Turn files	374	323	697
Usable turns	349	291	640 (91.82%)
Unusable turns (skipped when rendering)	25	32	57
Sessions with ≥ 1 unusable turn	–	–	40
Usable turns per session: median (mean; range)	4 (4.06; 1–9)	3 (3.38; 0–8)	7 (7.44; 1–17)
Sessions opened by this speaker (first usable turn)	85	1	–
Direction switches: total (per session: median; range)	–	–	540 (6; 0–14)
Same-speaker successions (skipped/adjacent)	–	–	14 (9/5)
<i>Audio</i>			
Usable turn audio before trimming (min)	24.563	28.068	52.631
Speech after trimming (min)	17.953	22.425	40.378
Untrimmed turn (s): mean/median/p90	4.223/3.64/7.02	5.787/4.76/10.94	4.934/4.0/9.064
Rendered turn (s): mean/median	3.086/2.42	4.624/3.38	3.785/2.86
Rendered turn (s): p90/max	5.492/12.64	9.92/17.98	7.76/17.98
Rendered turns longer than 5 s	–	–	144 (22.5%)
Rendered turns longer than 10 s	–	–	29 (4.5%)
Silence removed per turn (s): median leading/trailing	0.5/0.46	0.56/0.44	0.54/0.44
Speaking rate (words/s): trimmed (untrimmed) audio	2.997 (2.19)	2.666 (2.13)	–
<i>Text</i>			
Source words	3,228	3,587	6,815
Source words per turn: mean/median/p90/max	9.25/8/17/34	12.33/10/25/37	–
Reference words	3,339	3,278	6,617
Reference words per turn: mean/median/max	9.57/8/44	11.26/10/38	–
Vocabulary, source/reference (lowercased types)	472/381	534/602	–
Filled-pause marks [fil]	60	172	–
Noise labels (NOI tier)	235	228	–
Other marks: */~/**	53/8/1	61/4/11	–
Turns containing markup	101	135	–

Source and usable turns. BAS TAXI 2.5 (Bavarian Archive for Speech Signals (BAS), 2001), distributed through the BAS CLARIN repository (Reichel et al., 2016), holds telephone dialogues between a German-speaking taxi dispatcher and an English-speaking client, each speaking their own language. Every turn is a separate 8 kHz, 16-bit mono file. Speakers took turns with a push-to-talk button, which prevented overlap and kept no inter-turn timing, so only the order of the turns is known. Turns that passed the corpus validation carry a BAS Partitur annotation with orthographic words in SpeechDat notation (ORT tier), pronunciations (KAN), noise labels (NOI) and a human translation into the other language (TLN); turns rated as garbage (overlapping speech, meta talk, empty, noise only or wrong language) have none. We drop pre-dialogue and hang-up items, which leaves 697 dispatcher and client turn files in 86 sessions (IDs SES0037–SES0130, not contiguous), and keep a turn if its cleaned transcript is non-empty and it has a human translation whose direction matches the speaker’s language. Cleaning removes bracketed markers such as [fil] and unintelligible tokens (**), strips the marks of hardly recognisable (*) and truncated (~) words, and keeps word fragments. This leaves 640 usable turns (91.82%; German 349, English 291), exactly the turns that the validation list shipped with the corpus marks usable; 40 sessions contain at least

one unusable turn, which the renderer skips. The corpus README reports 645 usable turns from the original validation, and the corpus’s revalidation report counts 640, as we do; catalogue entries give 94 dialogues, the number recorded, but version 2.3 removed the eight that have no usable turn. The build identifies speakers only by role, so we report no speaker counts or demographics. [Draft: *add speaker counts, recording conditions and translation guidelines from the corpus documentation*]

Composition. German turns are shorter than English ones: after silence trimming, the median turn lasts 2.42 s in German and 3.38 s in English, and the median number of source words per turn is 8 and 10. The rendered turns amount to 40.378 min of speech after silence trimming (52.631 min of usable turn audio before trimming). Over both languages, the median turn lasts 2.86 s, 22.5% of the turns exceed 5 s and 4.5% exceed 10 s (Figure 10 breaks latency down by turn length). On trimmed audio, the speaking rate is 2.997 words/s in German and 2.666 words/s in English. The transcripts mark what synthetic speech lacks: 60 German and 172 English filled pauses and 8 and 4 truncated words, and the NOI tier holds 235 and 228 noise labels. The human translations are lowercase and unpunctuated, so the scorer normalises hypotheses to this style (Appendix C). Each direction has only about 3.3k reference words (3,339 English words for the German turns and 3,278 German words for the English turns), so every comparison uses session-clustered bootstrap intervals (Appendix H).

Rendering. The build resamples the usable turns from 8 to 16 kHz with a polyphase filter and keeps the telephone band; it applies no bandwidth extension, which matters when reading ASR error rates. It trims each turn to the span from the first to the last 20 ms frame whose RMS exceeds a threshold of 0.1 times the 95th-percentile frame RMS (but at least 10^{-4}), widened by a 0.1 s margin on each side. Trimming removes a median of 0.54 s of leading and 0.44 s of trailing silence per turn; the totals, 416.06 s and 319.14 s, account exactly for the difference between the untrimmed (3157.86 s) and the trimmed audio (2422.66 s). [Draft: *a median leading silence of 0.88 s (about 0.9 s in the code docstring) is not reproduced by this build; do not quote it*] Only the timing is sampled, as in simulated conversations for diarization training (Landini et al., 2022); the turns and their order are real. The renderer places the turns in dialogue order after a 0.5 s lead, separates consecutive turns of one speaker by a pause drawn from $\mathcal{U}[0.3, 0.8]$ s, appends a 0.5 s tail and sums the two speaker channels, clipped to $[-1, 1]$. At a speaker change, L0 natural draws the gap from $\mathcal{N}(0.2, 0.25^2)$ s clipped to $[0.05, 1.0]$ s, close to the typical gap of about 200 ms (Levinson & Torreira, 2015), which varies little across languages (Stivers et al., 2009); L0 mediated draws it from $\mathcal{U}[1, 3]$ s, for a listener who reads the translation before answering. At the 540 speaker changes, natural gaps have a mean of 0.257 s and a median of 0.223 s, and 129 of them (23.89%) sit at the 0.05 s floor, against about 27.4% expected from clipping the normal; mediated gaps have a mean of 1.995 s and a median of 2.026 s (Figure 6c). The renders last 0.737 h (natural; median session 29.023 s; speech occupancy 91.27%) and 0.998 h (mediated; 39.316 s; 67.42%), and the overlap ratio is 0.0 in all 86 sessions of both. Real conversation overlaps at many speaker changes (Heldner & Edlund, 2010); L1 therefore starts the next turn $\mathcal{U}[0.2, 0.8]$ s before the previous one ends at each speaker change, under its own configuration name and checksums. [Draft: *planned; the L1 renderer is not built yet*]

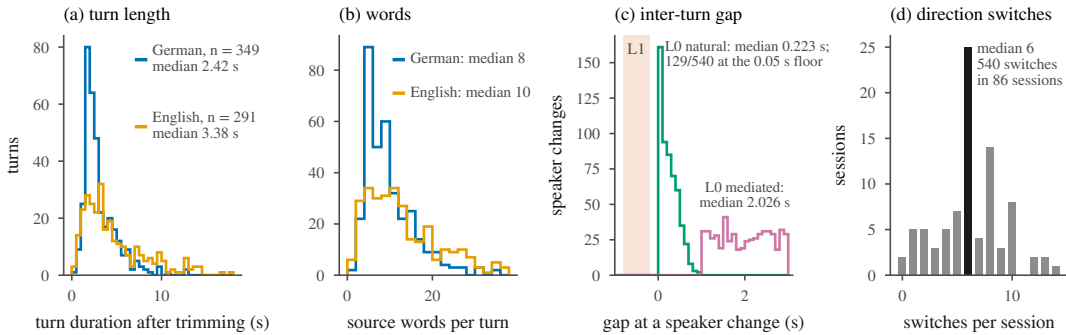


Figure 6. CST-Bench-TAXI statistics (86 sessions, 640 usable turns; aggregates only, as TAXI may not be redistributed). (a) Turn duration after silence trimming (0.1 s margins included), by source language. (b) Source words per turn. (c) Gaps drawn at the 540 speaker changes under L0 natural and L0 mediated; the shaded band of negative gaps marks where the planned L1 condition starts the next turn, 0.2–0.8 s before the previous turn ends. (d) Direction switches per session; the median (6) is drawn in black, and two sessions have no switch.

Caveats. Speech, content and turn order are real, but every gap is synthetic and the corpus has no overlap; overlap claims therefore rest on the L1 renders and on CST-Bench-Syn (Appendix B). Turn intervals include the 0.1 s margins, so the turn boundaries to which EndOffset and switch latency refer can be off by up to about 0.1 s, and energy trimming may clip

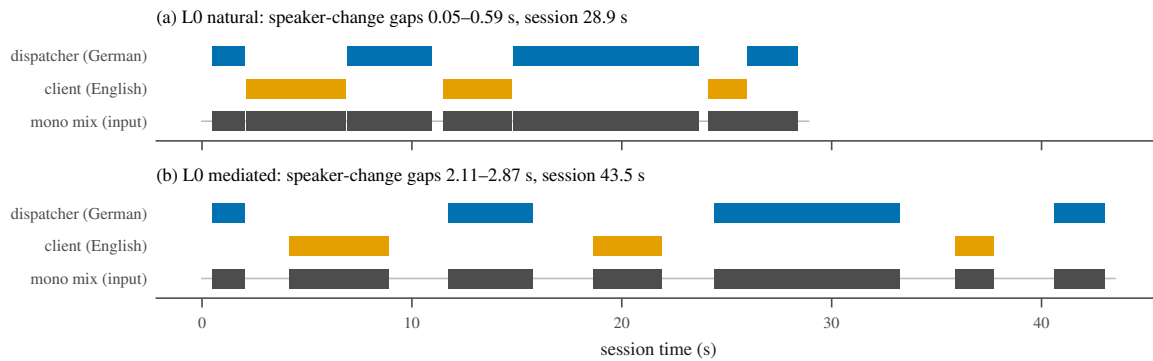


Figure 7. Session SES0104 (7 turns) rendered under (a) L0 natural (speaker-change gaps 0.05–0.59 s; session 28.9 s, close to the median session of 29.023 s) and (b) L0 mediated (gaps 2.11–2.87 s; session 43.5 s). Lanes show the German dispatcher, the English client and the mono mix, which is all a system receives. Speech and turn order are real; the timing is synthetic. Only turn intervals are shown, no audio or text.

soft onsets; forced-aligned word times will check these boundaries. *[Draft: planned; the TAXI turns are not force-aligned yet]* Language and role are confounded (German is always the dispatcher), and the dispatcher opens 85 of the 86 sessions, a trivial prior for the first direction. The realistic wrappers therefore route by spoken-language identification (Appendix D), and CST-Bench-Syn renders both speaker–language assignments. Sessions SES0085 and SES0095 hold a single usable turn, in German; they have no direction switch and are excluded from switch metrics. Fourteen turns follow a turn of the same speaker, 9 because unusable turns in between were skipped and 5 because the two turns are adjacent in the source; a pause separates them, and they do not count as switches. Finally, TAXI serves only for evaluation: no threshold, operating point or model is tuned on it.

Release and reproducibility. TAXI is free of charge under a scientific licence, and a commercial licence is available; redistribution, even in part, requires the written permission of the copyright holders. We therefore release session and turn IDs, configurations, build scripts, checksums and per-turn scores keyed by turn ID, but no audio, text or system output derived from TAXI (Table 19). Each rendered session directory holds `mix.wav` (the single-channel system input), `2ch.wav` (one channel per speaker, for analysis and oracle diagnostics only) and `timeline.json` (turn intervals with margins, gaps, removed silence, transcripts and references). The channel order of `2ch.wav` follows the `channels` field of the timeline: channel 0 is the English client and channel 1 the German dispatcher. *[Draft: the package README states the reverse order; fix it before release]* Per-session seeds derived from the session IDs and Python’s `random` module make the timelines identical on every platform (Section 4). `cstbench verify` requires the SHA-256 digest of every timeline (as sorted-key JSON) to match exactly and reports audio-hash differences separately, because audio hashes can depend on the `numpy`, `scipy` and `libsndfile` versions (the reference hashes were made with `numpy` 1.24.3, `scipy` 1.10.1 and `libsndfile` 1.2.2). A local rebuild matched 86/86 timeline digests and 86/86 `mix.wav` and `2ch.wav` hashes in both configurations. Released configurations are frozen, and any change gets a new name. With the corpus in `TAXI_DIR` (one folder per session), the track is built and verified by

```
cstbench build-taxi --root TAXI_DIR --out OUT_DIR
cstbench verify --out OUT_DIR/taxi-natural --reference checksums/taxi-natural.json
cstbench verify --out OUT_DIR/taxi-mediated --reference checksums/taxi-mediated.json
```

Why TAXI. A real CST test set needs one conversation between at least two real speakers who use different languages, with shared multi-turn context, per-turn audio, the identity and language of each speaker, turn order and transcripts; reference translations, turn timing, session audio and per-speaker channels are desirable. Parallel speech read from scripts, separately recorded sides of a text dialogue, merged monolingual corpora and utterance sets whose order or context cannot be recovered do not qualify. Table 8 checks the candidates against these requirements. MSLT (Federmann & Lewis, 2016; 2017) was recorded from bilingual calls, but the English side of each call was discarded. DRAL (Ward et al., 2023; 2024; Avila & Ward, 2023) and mDRAL (Seamless Communication et al., 2023) pair fragments of one-language conversations with re-enactments in the other language. InCroMin (Čechovič et al., 2025) records real meetings between people without a common language, but they talk through machine speech translation, each participant is recorded on a separate track, and no English–German pair is included. CS-Dialogue (Zhou et al., 2025) holds about 104 h of Mandarin–English code-switching without translations. The Verbmobil II interpreter volumes, from the Verbmobil project (Wahlster, 2000), contain German–English and German–Japanese dialogues and a human-interpreter subset but require a paid licence, and

the Fisher–CALLHOME Spanish–English corpus (Post et al., 2013) has a single source language. Among these candidates, only TAXI combines real two-language dialogues, human translations of both sides and free access for research. What it lacks, turn timing and overlap, is exactly what the renderer controls.

Table 8. Why CST-Bench-TAXI: real bilingual conversational speech resources checked against the requirements of a real CST test set (citations in the text of Appendix A). ✓ yes, ◦ partly, ✗ no; –: not assessed or not applicable.

Corpus	Same conversation, two languages	Real spontaneous speech	Human translations	Per-turn audio / timing	Overlap	Access / licence	Verdict
BAS TAXI 2.5	✓ German dispatcher, English client	✓	✓ both sides (TLN tier)	◦ per-turn files; no timing (push-to-talk)	✗ none	free scientific licence; no redistribution	used (real track)
MSLT	◦ bilingual calls; English side discarded	✓	◦ released side only	one side only	–	–	unsuitable (one-sided release)
DRAL / mDRAL	✗ one language per conversation; fragments re-enacted in the other	◦ originals only	◦ spoken re-enactments	paired fragments	–	LDC (DRAL)	unsuitable (re-enacted)
InCroMin	✓ meetings mediated by machine speech translation	✓	◦ corrected machine translations into English	per-participant tracks	–	CC BY 4.0	related work (12 languages, no English–German pair)
CS-Dialogue	◦ Mandarin–English code-switching within speakers	✓	✗	full dialogues	–	–	unsuitable (about 104 h, no translations)
Verbmobil II interpreter volumes	✓ German–English and German–Japanese; human-interpreter subset	✓	–	–	–	paid	not used
Fisher–CALLHOME Spanish–English ST	✗ one source language (Spanish)	✓	✓ crowd-sourced, into English	–	–	LDC	related (one source language)

B. CST-Bench-Syn Construction and Data Card

[Draft: planned: CST-Bench-Syn is not built yet; this appendix specifies its construction, and every measured value in Table 9 is pending]

CST-Bench-Syn is an evaluation-only set of synthetic (TTS) speech. It adds what CST-Bench-TAXI cannot provide: turn-end overlap that can be measured at the word level, a distant language pair, and both speaker–language assignments, which remove TAXI’s first-speaker prior and its tie between language and role. It is not real conversation and does not meet the requirements for a real test set (Appendix A). We therefore draw only relative conclusions from it (gaps between protocols and system rankings, never absolute scores across pairs) and test whether it ranks systems as CST-Bench-TAXI does (Table 11 and Figure 8). There is no training or development split: a calibration split of about 5% of the sessions, disjoint from the evaluation sessions, is the only data on which wrapper thresholds and operating points are chosen, with one rule for all systems. Table 9 is the data card.

Pairs and speaker assignment. En↔De is the main pair: it shares its languages with CST-Bench-TAXI, which makes the sim-to-real comparison possible. Ko↔En adds a distant word order. Every dialogue is rendered under both speaker–language assignments (speaker *A* speaks the first language and *B* the second, and vice versa), and the same dialogues serve both pairs, so differences between pairs come from the languages rather than from the content. We exclude German–Korean, which almost no baseline supports, and use English as the pivot.

Table 9. CST-Bench-Syn data card (planned). The set is not built yet: blank cells will hold measured values, and the last column gives plan targets, which are not results. A session is one dialogue under one speaker–language assignment; every dialogue is rendered under both. The re-recognition error is the WER for English and German and the CER for Korean; turns above the target are re-synthesised. MT: machine translation; n/a: not applicable; –: no target.

Statistic	En↔De		Ko↔En		Plan target
	XDailyDialog	BConTrasT	XDailyDialog	BConTrasT	
<i>Size</i>					
Dialogues / sessions					–
Turns per direction					–
Speech per timing condition (h)					–
Calibration split (% of sessions)					about 5
<i>Speech</i>					
Voices per language and edition					–
Re-recognition error (%)					≤ 5
Re-synthesised turns (%)					–
Speaker cosine, same / other speaker					–
TTS word times vs. forced alignment					–
<i>Text</i>					
Reference type	human	human	Ko: consensus MT En: human	Ko: consensus MT En: human	–
Korean consensus pass rate (%)	n/a	n/a			≥ 85
Korean spot-check error rate (%)	n/a	n/a			–
Contamination-probe agreement					–
Licence	CC BY-NC-SA 4.0	CC BY-SA 4.0	CC BY-NC-SA 4.0	CC BY-SA 4.0	–

Dialogues. XDailyDialog (Liu et al., 2023) provides professional translations of the everyday dialogues of DailyDialog (Li et al., 2017), and BConTrasT (Farajian et al., 2020) provides professional English–German translations of task-oriented Taskmaster-1 dialogues between a customer and an agent in six domains. Speaker *A*’s turns come from one language version of a dialogue and *B*’s from the other, so both share one context and the parallel version of each turn is its reference. We select dialogues of 6–16 turns and balance the domains, the turn lengths and the share of dialogues that contain numbers or named entities. XDailyDialog turns are short (about 11.5 tokens on average), so the selection favours multi-sentence turns, which leave room for turn-end overlap. No parallel corpus that cannot be redistributed is used.

Contamination. DailyDialog and Taskmaster-1 text may be in the pretraining data of the LLMs that serve as baselines and teachers. For each of them, a probe gives the opening turns of a selected dialogue, lets the model continue it and measures the agreement of the continuation with the original (Table 9). Generation prompts, models and seeds are released. No DailyDialog, Taskmaster-1 or TAXI text and no CST-Bench-Syn dialogue or voice is used to train MODEL (Section 5).

References. English–German references are the human translations of the sources. Neither source contains Korean, so every Korean text, both the Korean turns that are synthesised and the Korean references of English turns, is translated from English by three independent systems: Qwen3.8-27B, MADLAD-400-10B-MT and EXAONE-4.0. A text passes if at least two of the three translations agree, as judged by MetricX-24 in its reference-free mode (Juraska et al., 2024), CometKiwi (Rei et al., 2022b), the LaBSE similarity of back-translations, and matching numbers and named entities; text that fails is removed, and we report the pass rate. The released Korean text is the MADLAD-400 translation, as no evaluated system belongs to that model family: Qwen3.8-27B is also a baseline translator and the teacher of our commit targets (Section 5), and EXAONE-4.0, whose licence restricts its outputs to non-commercial use, serves only in the agreement check, so its outputs are neither released nor used for training. [Draft: fix the rule for text whose MADLAD-400 translation is the dissenting one] Native Korean speakers annotate the errors in a fixed random sample with a light MQM-style scheme. Ko↔En enters the main results only if at least 85% of the Korean text passes and the speech meets the re-recognition target below; otherwise it is reported as a pilot. For En↔De, silver references built in the same way test whether gold and silver references rank systems alike (Table 11). Gold and silver segments are aggregated separately, and reference-free CometKiwi is reported next to the reference-based metrics.

Speech synthesis and quality assurance. A commercial edition carries the main results: Azure neural TTS (Microsoft, 2026) with native preset voices for en-US, de-DE and ko-KR, balanced by gender, and word boundaries from the synthesiser. An open edition, CosyVoice 3 (Du et al., 2025) or Qwen3-TTS (Hu et al., 2026) (both with Apache-2.0 weights), covers a subset and tests whether the TTS engine changes system rankings (Table 11). A pilot of 50 sentences per language admits an engine only if its re-recognition error is at most 5%. Each speaker keeps one voice throughout a dialogue. Voices are presets, designed voices or voices cloned from CC0 reference recordings; no real person’s voice is cloned without consent, and no voice is cloned across languages. Every synthesised turn is re-recognised with Whisper-large-v3 (Radford et al., 2023) and Qwen3-ASR (Shi et al., 2026); turns with a WER (CER for Korean) above 5% are re-synthesised, and we report the error and the re-synthesis rate. Speaker consistency is measured by the cosine similarity of speaker embeddings within and between speakers. The synthesiser’s word boundaries are compared with forced alignment (Qwen3-ForcedAligner; Shi et al., 2026), and we report the distribution of their differences, because word-level overlap and the reference points of latency depend on these times.

Rendering and release. Sessions are rendered with the same renderer and rules as CST-Bench-TAXI (Section 4) under L0 natural and L1, optionally also under L0 mediated, each with its own configuration name and checksums; overlap is measured on turn intervals and on word boundaries. Mixes are clean: noise and reverberation variants are outside this release, and a telephone-band variant is added only if CST-Bench-Syn and CST-Bench-TAXI rank systems differently. Each session comes with the mono mix, per-speaker channels (for analysis and oracle diagnostics only), a timeline with turn and word times, and references flagged as gold or silver. A data card discloses that the speech is synthetic and lists the sources, licences, generation models, prompts, seeds and QA results. CC BY-NC-SA 4.0 and CC BY-SA 4.0 are incompatible ShareAlike licences, so the XDailyDialog-based and the BConTrasT-based sessions form two subsets, each under the licence of its source (Taskmaster-1, on which BConTrasT builds, is CC BY 4.0). The commercial edition is synthesised only on a paid tier (Table 19). Instead of a fixed size in hours, CST-Bench-Syn is sized by a power estimate from the variance of TAXI segment scores for the pre-registered comparisons: a first tranche is built and extended only if its confidence intervals are too wide.

Out of scope. This release leaves out the parts of an earlier design that targeted interruptions and barge-in: episodes of two to four history turns, an interruption event and one or two resolution turns; information-causal event generation with three difficulty levels; sizing by at least 300 events per combination of event type, difficulty and pair, which implied about 10–12 h per pair; partial references for interrupted turns; leakage checks for generated events; and a high-overlap condition with at least 25% overlap. Interruptions remain future work (Section 7).

Limitations. TTS prosody lacks disfluencies, reactions to the partner and the Lombard effect; the dialogue text is largely planned rather than spontaneous; one side of every pair is translationese; and the Korean references are silver. Rank agreement with CST-Bench-TAXI and between the two TTS editions (Table 11) indicates how far conclusions drawn on CST-Bench-Syn transfer to real speech.

C. Scorer Specification and Metric Validation

This appendix specifies `cstbench eval` precisely enough to reimplement it and describes how we test that its metrics measure what they are meant to measure. As in Sections 3 and 4, turn n of a session of duration T has speaker s_n , interval $[b_n, e_n]$, duration $L_n = e_n - b_n$ and a reference translation r_n of $|r_n|$ words. *[Draft: the released scorer implements the submission format, normalisation, resegmentation, oracle mode, BLEU, chrF, StreamLAAL, EndOffset, empty turns and wrong-language words given gold turn IDs; COMET, switch latency, the wrong-direction detector, the latency variants, the bootstrap and all transcript and speaker metrics are planned]*

C.1. Submission, clocks and normalisation

Submission. A system submits one JSON line per committed piece with four fields: session ID, output language, emission time t in seconds from the start of `mix.wav`, and text. The output language selects the direction (English output translates German speech), so no speaker or direction label is needed. Pieces are never retracted, and every line counts as committed. The scorer sorts the pieces of each session by emission time, keeping file order for ties, splits each normalised piece into words and gives every word the time of its piece. An optional gold turn ID per piece selects the oracle mode described below. Further fields are kept, and any of them can serve as the clock; pieces of unknown sessions are reported

and ignored.

Clocks. On the ideal clock, t is the audio consumed when the piece is committed; the computation-aware clock (Ma et al., 2020c) adds processing time measured on fixed hardware (Table 14). Unlike SimulEval (Ma et al., 2020a), which runs a policy on each source segment, `cstbench eval` sees only committed pieces and their times, so they can come from any streaming harness, such as `simulstream` (Gaido et al., 2025), or from a unified model. On CST-Bench-TAXI, turn intervals include the 0.1 s margins of energy trimming (Appendix A), which can move the reference points of every latency metric by up to about 0.1 s.

Normalisation. Hypotheses and references are normalised alike, to the style of the TAXI references (lowercase, unpunctuated, numbers in words): (i) Unicode NFC; (ii) each number, i.e., a digit string with an optional decimal part, is spelled out with `num2words` in the output language, reading a decimal comma as a decimal point in German and dropping commas as thousands separators in the other languages; (iii) lowercasing; (iv) every character other than a letter, a digit, whitespace or a word-internal apostrophe becomes a space; (v) splitting at whitespace. The `num2words` version is pinned, because without the package step (ii) would be skipped silently. *[Draft: the released scorer skips step (ii) silently if num2words is missing]*

Quality. Per direction, BLEU (Papineni et al., 2002) and chrF (Popović, 2015) are sacreBLEU corpus scores (Post, 2018) over the turns of that direction, with default settings except for sacreBLEU’s Korean tokenizer for BLEU into Korean; we report the signatures and two decimals. An empty turn enters as an empty hypothesis and lowers both scores. COMET with the `wmt22-comet-da` model (Rei et al., 2020; 2022a) scores the same resegmented turns, with the gold transcript as the source. A per-turn file holds each turn’s source, hypothesis, reference and latencies; for CST-Bench-TAXI, only per-turn scores keyed by turn ID are released (Appendix H). *[Draft: planned: COMET inside the scorer and the Korean tokenizer; the released scorer leaves COMET to external tools reading the per-turn file]*

C.2. Resegmentation and oracle mode

For each session and output language λ , `cstbench eval` assigns the hypothesis words to the turns $\mathcal{N}_\lambda$ that are translated into λ , with the minimum-edit-distance criterion of `mwerSegmenter` (Matusov et al., 2005) as used in stream-level evaluation (Iranzo-Sánchez et al., 2021; Papi et al., 2024):

1. order the hypothesis words $w_1, \dots, w_M$ by emission time;
2. concatenate the normalised references r_n , $n \in \mathcal{N}_\lambda$, in dialogue order into $u_1, \dots, u_K$, and record the turn $o(k)$ of each reference word u_k ;
3. align $w_{1:M}$ to $u_{1:K}$ by word-level Levenshtein alignment with unit costs; on ties, the backtrace prefers a match or substitution to an insertion, and an insertion to a deletion;
4. assign each w_i aligned to some u_k , as a match or a substitution, to turn $o(k)$; assign each inserted word to the turn of the preceding aligned word or, if it precedes all aligned words, to the turn of the first aligned word.

Turn n thus receives the words h_n , with emission times $\tau_{n,1} \leq \dots \leq \tau_{n,|h_n|}$, and is empty if h_n is empty. Resegmentation looks only at text, so a word can land in a turn that begins after the word was emitted, and latency on resegmented output inherits a segmentation-related bias (Polák et al., 2025).

Oracle mode. If every piece of a session and direction carries a gold turn ID, the scorer skips resegmentation and assigns words to their turns directly; words whose ID is not a turn of direction λ count as wrong-language words and are not scored. This mode is only a diagnostic (Section 3): scoring the same output both ways measures how much resegmentation alone moves the scores.

C.3. Latency

For a non-empty turn n , let $d_{n,j} = \max(0, \tau_{n,j} - b_n)$ be the delay of its j -th word from the turn start, $\rho_n = L_n / \max(|h_n|, |r_n|)$ the rate, and J_n the first j with $d_{n,j} \geq L_n$, or $|h_n|$ if there is none. Then

$$\text{StreamLAAL}_n = \frac{1}{J_n} \sum_{j=1}^{J_n} (d_{n,j} - (j-1)\rho_n), \quad (7)$$

$$\text{EndOffset}_n = \tau_{n,|h_n|} - e_n, \quad (8)$$

$$\text{Switch}_n = d_{n,1} \quad \text{for } n > 1 \text{ with } s_n \neq s_{n-1}. \quad (9)$$

Equation (7) is LAAL (Papi et al., 2022), the length-adaptive form of Average Lagging (Ma et al., 2019), applied to resegmented turns, i.e., StreamLAAL (Papi et al., 2024). Two cases show what the metrics separate. If a turn’s translation is committed at e_n , every delay equals L_n , so $J_n = 1$, $\text{StreamLAAL}_n = L_n$ and $\text{EndOffset}_n = 0$: offline systems on oracle turns score the mean turn duration, 4.624 s for En→De and 3.086 s for De→En on CST-Bench-TAXI. If instead the j -th of the $|r_n|$ reference words is committed at $b_n + jL_n/|r_n|$, then $\text{StreamLAAL}_n = L_n/|r_n|$, the duration of one word, while EndOffset_n is still 0. StreamLAAL thus measures the lag within a turn and EndOffset the wait after it. EndOffset is negative if a turn’s output ends before the speaker does, through anticipation or truncation, so we always report it with quality and length ratio. Switch latency (Equation (9)) is undefined for the first turn of a session, so the two single-turn CST-Bench-TAXI sessions contribute none.

Aggregation. Per direction, we report the mean, median and 90th percentile over non-empty turns, to three decimals. The median averages the two middle values of an even count, and the 90th percentile of m values is the sorted value at the 0-based index $\min(m-1, \lfloor 0.9m \rfloor)$. The empty-turn rate is reported next to every latency.

Penalised variant. Averaging over non-empty turns lets a system improve its latency by dropping hard turns. The penalised variant therefore scores an empty turn as if its whole reference were committed at the session end: $\text{EndOffset}_n = T - e_n$ and $\text{StreamLAAL}_n = \text{Switch}_n = T - b_n$. On the ideal clock no piece comes later than T , so these are the worst values any output can obtain, and dropping a turn never lowers its latency.

Exactly matched words. Unrelated output can lower StreamLAAL (row 2 of Table 10); one mechanism is that a word assigned to a turn that begins after the word was emitted gets delay 0. We therefore also report Equations (7) and (8) with h_n restricted to the words that exactly match their aligned reference word, so that unrelated words cannot contribute. [Draft: planned: switch latency, the penalised variant and the matched-word variant]

C.4. Errors and auxiliary outputs

Empty turns and length ratio. A turn is empty if no word is assigned to it. Per direction, we report the share of empty turns and the ratio of hypothesis to reference words.

Wrong-direction words. In oracle mode, wrong-direction words are the wrong-language words defined above. Without turn IDs, a piece (t_i, λ_i, y_i) is checked against the turns open at its emission time, $\mathcal{A}(t_i) = \{n : b_n \leq t_i \leq e_n + W\}$, with a window W fixed before any system is scored. All words of the piece are wrong-direction words if (a) *wrong channel*: $\mathcal{A}(t_i)$ is non-empty and all its turns are spoken in λ_i , so no open turn needs a translation into λ_i ; (b) *wrong language*: a text language identifier restricted to the session’s two languages assigns y_i to the language other than λ_i ; or (c) *copy-through*: y_i is closer to the transcript of an open turn spoken in λ_i than to the reference of every open turn translated into λ_i , where the distance from y_i to a text is the smallest word edit distance between y_i and a span of that text, divided by the length of y_i . Under turn-end overlap, turns in both languages are open at once, so (a) cannot fire and (b) and (c) decide. Rows 2 and 8 of Table 10 test (b) and (c), and the delayed rows check that late but correct output is not flagged. We report wrong-direction words as a percentage of all committed words and compare the detector with the oracle-mode count wherever turn IDs exist. [Draft: planned: the detector, its window W and the text language identifier; the released scorer counts wrong-language words only given gold turn IDs]

Transcripts and speakers. For systems with a transcript channel, we report WER per language after the same normalisation; cpWER (Watanabe et al., 2020), which compares each speaker’s concatenated words under the best mapping

Table 10. Metric meta-evaluation on CST-Bench-TAXI, L0 natural: controlled perturbations of the reference translations, with the change expected for each metric fixed in advance (construction of each row in Appendix C). Q: BLEU and chrF; L: mean StreamLAAL (s); E: mean EndOffset (s), both over non-empty turns; W: number of wrong-direction words; Δ : added delay; $L_n, |r_n|$: duration and reference length of turn n ; $=$: unchanged; $\uparrow/\downarrow$: increases/decreases. [†]Preliminary; to be re-run under the final protocol. *The current scorer counts such words only given gold turn IDs. Blank cells are pending.

Perturbation	Expected change	En→De			De→En			W
		BLEU/chrF	L	E	BLEU/chrF	L	E	
1 Reference at turn end	Q = 100; L = mean L_n ; E = 0; W = 0	100.0/100.0 [†]	4.624 [†]	0.0 [†]	100.0/100.0 [†]	3.086 [†]	0.0 [†]	0 [†]
2 Source transcript passed through untranslated	Q $\approx$ 0; L, E not below row 1; W = all words	2.38/27.34 [†]	4.366 [†]	2.576 [†]	0.39/21.86 [†]	2.276 [†]	1.421 [†]	0 ^{†*}
3 Row 1 delayed by 0.5 s	Q =; L, E + Δ ; W = 0							
4 Row 1 delayed by 1 s	Q =; L, E + Δ ; W = 0							
5 Row 1 delayed by 2 s	Q =; L, E + Δ ; W = 0							
6 Reference spread evenly over the turn	Q =; L = mean $L_n/ r_n $; E = 0; W = 0							
7 Row 6 without the last third of each turn	vs. row 6: Q $\downarrow$; L =; E $\downarrow$ (below 0); W = 0							
8 Turns merged across a speaker change (VAD)	Q $\downarrow$; E $\uparrow$; W $\uparrow$; more empty turns							
9 Output of 10% of turns dropped	Q $\downarrow$; L, E $\approx$ row 1; 10% empty turns							
10 Rows 1, 3–7, 9 with gold turn IDs: largest change	0 for every metric							

between hypothesis and reference speakers, so that arrival-order labels need no fixed identity; and its time-constrained variant tcpWER from MeetEval (von Neumann et al., 2023). DER uses collars of 0 and 0.25 s at 80 ms resolution. [Draft: planned]

Speaker-attributed BLEU. DiariST scores translations per speaker after mapping hypothesis speakers to reference speakers (Yang et al., 2024). In CST each speaker uses one language, so the reference translations of speaker A are exactly those of direction $\ell_A \rightarrow \ell_B$. For a system that labels its translations with speakers, speaker-attributed BLEU therefore scores the same hypothesis–reference pairs as per-direction BLEU whenever every piece labelled A is in ℓ_B and every piece labelled B is in ℓ_A . The two differ only on pieces whose label contradicts their language; these are attribution or direction errors, which the wrong-direction rate and cpWER already count. We therefore compute speaker-attributed BLEU only as a consistency check for systems with speaker labels.

C.5. Metric meta-evaluation

We perturb the reference output on CST-Bench-TAXI L0 natural and check that each metric moves as expected while the others stay put (Table 10); the expected changes are fixed in advance. Row 1 commits every reference translation at its turn end. Row 2 emits each turn’s source transcript as its translation, as a system that passes speech through untranslated would. Rows 3–5 delay row 1 by Δ . Row 6 commits the j -th reference word of turn n at $b_n + jL_n/|r_n|$, and row 7 drops the last $\lfloor |r_n|/3 \rfloor$ words of every turn of row 6. Row 8 simulates a VAD that merges two turns across a speaker change, at a seeded subset of speaker changes: the first turn’s reference is committed at the end of the second turn, followed on the same channel by the second turn’s source transcript, as if the merged segment had been routed by the first turn’s language. Row 9 removes the output of a seeded 10% of turns. Row 10 rescores rows 1, 3–7 and 9, whose words all come from the right turns, with gold turn IDs and reports the largest change. No final system result is reported until every row behaves as expected or the deviation is explained; the preliminary system results have not yet passed this check and will be re-run. We repeat the comparison of row 10 for every system run under the oracle wrapper, whose pieces carry turn IDs.

Rows 1 and 2 exist so far only as preliminary runs ([†]; to be re-run under the final protocol). Row 1 behaves as derived above: BLEU and chrF are 100.0[†], EndOffset is 0.0[†] s, no word counts as wrong, and the mean StreamLAAL equals the mean turn duration (4.624[†] s for En→De, 3.086[†] s for De→En). Row 2 drives BLEU down to 2.38[†] for En→De and 0.39[†] for De→En, while chrF keeps 27.34[†] and 21.86[†] from shared character n-grams. However, it *lowers* the mean StreamLAAL below that of the reference (4.366[†] vs. 4.624[†] s for En→De; 2.276[†] vs. 3.086[†] s for De→En) and raises the mean EndOffset

Table 11. Validity of system rankings (TAXI: CST-Bench-TAXI; Syn: CST-Bench-Syn; all L0 natural). Each row ranks the same n configurations (system $\times$ wrapper $\times$ operating point $\times$ direction) in two ways and reports Kendall’s τ_b with a 95% CI from 2,000 session-bootstrap resamples. For render seeds, τ_b is the smallest pairwise value among the three renders; Appendix C also compares their standard deviation with the CI width. Blank cells are pending.

Comparison	Configurations compared	n	Kendall’s τ_b	95% CI
<i>Speech, construction and rendering (chrF rankings)</i>				
TAXI vs. Syn En $\leftrightarrow$ De	those run on both sets			
Commercial vs. open TTS edition	Syn subset			
Gold vs. silver references	Syn En $\leftrightarrow$ De			
Render seeds 1–3 (chrF and EndOffset)	TAXI			
<i>Metric choice</i>				
StreamLAAL vs. EndOffset	TAXI			
StreamLAAL vs. LongYAAL	TAXI			
COMET vs. chrF	TAXI			

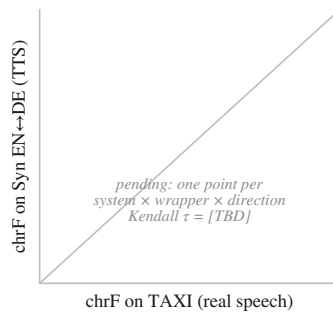


Figure 8. Sim-to-real agreement: chrF of each system, wrapper and direction at its main operating point on CST-Bench-TAXI (real speech) and on CST-Bench-Syn En $\leftrightarrow$ De (TTS), both under L0 natural. Hollow markers: realistic wrapper; grey line: identity. Kendall’s τ_b is annotated in the panel; its bootstrap CI is in Table 11. [Draft: placeholder; the points appear once the baselines have run on both sets]

to 2.576[†] and 1.421[†] s, presumably because resegmentation spreads the untranslated words over neighbouring turns; and without turn IDs, the current scorer counts none of these words as wrong. These two failures motivate the matched-word latency, the rule to show latency only next to quality, length ratio and empty turns, and the wrong-direction detector. In addition, five unit tests check normalisation, resegmentation, LAAL, oracle versus resegmented scoring and the command-line interface on toy sessions; for instance, an oracle hypothesis scores BLEU 100.0 with a mean EndOffset of 0.0 s, and a two-word turn reaches chrF 100.0. [Draft: re-run rows 1–2 under the final protocol and record how the row-2 hypothesis is time-stamped]

C.6. Rank agreement and sim-to-real validity

Several choices could change our conclusions without changing any system: synthetic instead of real speech, the TTS engine, silver instead of human references, the random render, and the metric. Table 11 measures each by Kendall’s τ_b between two rankings of the same configurations (system, wrapper, operating point and direction), with a 95% CI from 2,000 bootstrap resamples of sessions, drawn independently in each set when two sets are compared (Appendix H).

Synthetic versus real speech. The main check ranks all configurations that run on both sets by chrF on CST-Bench-TAXI and on CST-Bench-Syn En $\leftrightarrow$ De, both under L0 natural, so that only the speech and the dialogues differ (Figure 8). Before the runs, we fix a threshold τ^* ([TBD]): if the lower CI bound reaches τ^* , CST-Bench-Syn may rank systems on its own; otherwise we report the disagreement as a finding, use CST-Bench-Syn only for contrasts within one set, such as oracle versus realistic wrappers or L0 versus L1, and test whether a telephone-band rendering of CST-Bench-Syn restores agreement. The commercial and the open TTS edition are compared in the same way on a subset of CST-Bench-Syn.

References and renders. To test how far silver references can rank systems, we rescore CST-Bench-Syn En $\leftrightarrow$ De against silver references produced like the Korean ones (Appendix B) instead of the human references. To test render noise, we

re-render CST-Bench-TAXI L0 natural with seeds 1–3 under new configuration names and compare the standard deviation of chrF and EndOffset across seeds with the bootstrap CI width; Table 11 lists the smallest pairwise τ_b among the three renders.

Metrics. StreamLAAL and EndOffset measure different waits, so a low τ_b between them is a finding rather than a validity failure: it shows that the listener’s wait reorders systems. LongYAAL (Polák et al., 2025), computed on SoftSegmenter resegmentation in OmniSTEval, was the primary latency metric of IWSLT 2026, which reported StreamLAAL for comparison (Adelani et al., 2026); we report how well the two agree. Rankings by string metrics alone can mislead (Kocmi et al., 2021), which favours COMET, but the TAXI references are lowercase and unpunctuated. COMET is therefore primary on CST-Bench-TAXI only if the lower CI bound of τ_b between its ranking and that of chrF reaches τ^* ; otherwise chrF is. *[Draft: planned: every row of Table 11 and the points of Figure 8]*

D. Baselines and Wrappers

Table 12 lists the systems of Section 6. All of them are scored by `cstbench eval` from committed, time-stamped pieces (Appendix C); this appendix describes how each system turns a session into pieces, how its latency settings are chosen, and which prompts the LLM-based systems use. We use public checkpoints without fine-tuning on CST data; the only tuning data is the CST-Bench-Syn calibration split, and nothing is tuned on CST-Bench-TAXI. Seed LiveInterpret 2.0 (Cheng et al., 2025) is closed and covers only Chinese–English, so we discuss it but do not run it. *[Draft: implemented: the offline upper bounds, whose CST-Bench-TAXI scores are preliminary; planned: the streaming bound, both wrappers, the streaming baselines and the operating points]*

Offline upper bounds. Each gold turn is cut from `mix.wav` at its reference interval, 0.1 s margins included, and translated after it ends, one turn per prompt and without dialogue history. Qwen3-ASR-1.7B (Shi et al., 2026) transcribes the turn with its language given, and Qwen3.8-27B translates this transcript, or the gold transcript, from the user message

```
Translate this {src_lang} utterance from a phone conversation into {tgt_lang}.
Output only the {tgt_lang} translation.
```

followed by a blank line and the turn’s text; the placeholders are English names of the languages, such as German. Decoding is greedy with at most 160 new tokens and the model’s reasoning mode switched off, and the first line of the answer is kept. Whisper-large-v3 (Radford et al., 2023) translates German turns, and on CST-Bench-Syn Korean turns, into English with its translation task and the source language given; it cannot translate into German. All models process batches of 16 turns. Every piece carries its turn ID and is stamped at the turn end e_n , so on the ideal clock EndOffset is 0 and StreamLAAL equals L_n (Appendix C). The computation-aware clock adds the per-turn compute, measured as the time of a batch divided by its size and therefore approximate; Table 14 re-measures it without batching. The WER of the ASR transcripts is computed after the scorer’s normalisation.

Streaming bound. To bound what the commit policy costs and saves when recognition and segmentation are perfect, gold transcripts with forced-aligned word times (Qwen3-ForcedAligner; Shi et al., 2026) are revealed word by word to Qwen3.8-27B, which commits translations at verified meaning units (MU2; Section 5), accepting a unit if its chrF against the aligned part of the full translation is at least 50. A unit is stamped when its last source word ends, and the rest of a turn is flushed at the turn end. *[Draft: planned; the German minimum unit is set on held-out data before the run]*

Oracle wrapper. A single-direction system runs as two instances, one per target language. The wrapper cuts each gold turn $[b_n, e_n]$ from the mono mix, so that under L1 the cut also contains the overlapping speech of the other speaker, and streams it in real time to the instance that translates into the other speaker’s language. Instances are reset at every turn and receive no dialogue history. A piece emitted after the instance has consumed u seconds of the turn is stamped $b_n + u$ on the ideal clock, plus compute time on the computation-aware clock, and carries the turn ID, which the scorer uses only for diagnostics. Cutting the turn from its speaker’s channel of `2ch.wav` instead, in the channel order recorded in the timeline, gives the oracle-separation variant of Table 4. *[Draft: planned]*

Realistic wrapper. The realistic wrapper receives only the mix and the language pair. Silero VAD (Silero Team, 2024) segments the stream online: a segment opens at a detected speech onset and closes after a minimum silence. A spoken-language identifier restricted to the session’s two languages labels each segment from its onset and updates the label as

Table 12. Baseline configurations, named as in Tables 3 and 6 (LLM: Qwen3.8-27B). Params: recognition plus translation models (B: billions; MODEL: total; its encoder comes from Nemotron 3.5 ASR Streaming 0.6B, which has 600M parameters per its model card and 638M as measured, RNN-T decoder included). Directions: En $\leftrightarrow$ De on CST-Bench-TAXI and, for systems with a Korean entry, on CST-Bench-Syn; Ko $\leftrightarrow$ En on CST-Bench-Syn only. Policy: when text is committed; LA-2: LocalAgreement-2. Latency knob: the setting swept on the CST-Bench-Syn calibration split; its two values (low and high latency) are chosen by the rule of Appendix D and are pending. Wrappers: O oracle segmentation, R realistic (VAD and language-ID routing), – none (the system handles the whole mix). Not listed: the diarization cascades of Table 5, which do not translate (Appendix D). *Optional; InfiniSST only if its weights are public. Blank cells are pending.

System	Checkpoint / version	Params	Directions	Policy	Latency knob (two points)	Wrappers	Licence
<i>Offline upper bounds: gold turns, each translated after it ends</i>							
Gold $\rightarrow$ LLM	Qwen3.8-27B	27B	En $\leftrightarrow$ De Ko $\leftrightarrow$ En	turn end	–	O	Apache-2.0
ASR $\rightarrow$ LLM	Qwen/Qwen3-ASR-1.7B; Qwen3.8-27B	1.7B + 27B	En $\leftrightarrow$ De Ko $\leftrightarrow$ En	turn end	–	O	Apache-2.0
Whisper ST	openai/whisper-large-v3		De $\rightarrow$ En Ko $\rightarrow$ En	turn end	–	O	
<i>Streaming bound: gold transcripts with forced-aligned word times</i>							
Gold $\rightarrow$ LLM (MU2)	Qwen3.8-27B	27B	En $\leftrightarrow$ De	MU2 units	one point (chrF $\geq$ 50)	O	Apache-2.0
<i>Pipelines built from single-direction systems</i>							
Consecutive cascade	Silero VAD; LID; Qwen3-ASR-1.7B; Qwen3.8-27B	1.7B + 27B	En $\leftrightarrow$ De Ko $\leftrightarrow$ En	VAD endpoint	endpoint silence	R	
SeamlessStreaming			En $\leftrightarrow$ De Ko $\leftrightarrow$ En	EMMA	decision threshold	O, R	
2 $\times$ SeamlessStreaming	as above		En $\leftrightarrow$ De	EMMA	as above	–	
Streaming cascade	nvidia/nemotron-3.5- asr-streaming-0.6b; Qwen3.8-27B	0.6B + 27B	En $\leftrightarrow$ De Ko $\leftrightarrow$ En	LA-2	ASR chunk, update step	O, R	OpenMDW-1.1; Apache-2.0
M4T v2 + AlignAtt	SeamlessM4T v2		En $\leftrightarrow$ De	AlignAtt	frames f	O, R	
Canary-1B-v2 + AlignAtt*	Canary-1B-v2	1B	En $\leftrightarrow$ De	AlignAtt	frames f	O, R	
StreamSpeech			De $\rightarrow$ En	learned	chunk size	O, R	
InfiniSST*			En $\rightarrow$ De	learned (LLM)		O, R	
<i>Cascades in the ablations (Table 6)</i>							
Same-backbone cascade (M1 $\rightarrow$ LLM)	MODEL M1; Qwen3.8-27B	2,336M + 27B	En $\leftrightarrow$ De	LA-2	update step	–	OpenMDW-1.1; Apache-2.0
Size-matched cascade (ASR $\rightarrow$ 1.7B LLM)	Qwen3-ASR-1.7B; LLM of $\approx$ 1.7B	1.7B + $\approx$ 1.7B	En $\leftrightarrow$ De	LA-2	update step	R	
<i>Ours</i>							
MODEL	Nemotron 3.5 encoder; Qwen3-ASR-1.7B thinker	2,336M	En $\leftrightarrow$ De Ko $\leftrightarrow$ En	MU2 units, delay δ_t	δ_t	–	OpenMDW-1.1; Apache-2.0

audio arrives. The segment is streamed to the instance that translates into the other language and is re-routed to the other instance if the label flips; text already committed stays. Pieces are stamped from the segment onset as in the oracle wrapper. As identifier we use the automatic language detection of Nemotron 3.5 ASR (NVIDIA, 2026), or the language identification of Qwen3-ASR (Shi et al., 2026) if the former proves less accurate on the calibration split. The VAD’s speech-probability threshold, minimum silence and speech padding and the identifier’s settings are fixed once on that split, for all systems. A short minimum silence splits turns at pauses, whereas a long one merges turns across the short L0 natural gaps (median 0.223 s on CST-Bench-TAXI); under L1, the VAD hears one continuous segment across the speaker change, so the second turn inherits the first turn’s route unless the label flips. Sweeps of the minimum silence serve analysis only, not tuning. The factorial of Table 4 combines components of the two wrappers: gold turns with the identifier, and VAD segments with the gold language of the turn that overlaps each segment most. *[Draft: planned; the identifier is chosen on the calibration split]*

Streaming systems. SeamlessStreaming (Seamless Communication et al., 2023) translates speech into a requested language with Efficient Monotonic Multihead Attention (EMMA), a variant of monotonic multihead attention (Ma et al., 2020b), and runs behind both wrappers. Two SeamlessStreaming instances on the whole mix, one per target language and without routing, form the naive bidirectional baseline: each instance also receives speech that is already in its target language, and what it emits for that speech counts as wrong-direction output (copy-through or wrong language; Appendix C). The streaming cascade passes the growing transcript of Nemotron 3.5 ASR (NVIDIA, 2026), a cache-aware streaming RNN-T, to Qwen3.8-27B, which re-translates it after every update, with the committed translation as a forced prefix and the instruction of the offline prompt, and commits the longest common prefix of its last two translations (LocalAgreement-2; Liu et al., 2020; Polák et al., 2022; Macháček et al., 2023). It follows recent cascades that translate a growing transcript with an LLM (Fuxa & Macháček, 2026), but its policy needs no access to attention weights. SeamlessM4T v2 (Seamless Communication et al., 2023; SEAMLESS Communication Team et al., 2025) and, optionally, Canary-1B-v2 (Sekoyan et al., 2025) are offline speech translation models that we run simultaneously in simulstream (Gaido et al., 2025) with AlignAtt (Papi et al., 2023a), which emits a token only if the audio frame it attends to most is not among the last f frames. StreamSpeech (Zhang et al., 2024), which learns translation and policy jointly, covers De→En; InfiniSST (Ouyang et al., 2025), an LLM-based system for unbounded speech, covers En→De and is included only if its weights are public. *[Draft: planned; checkpoints and parameter counts left blank in Table 12 are filled when the runs are set up]*

Cascades. The consecutive cascade is the realistic counterpart of the offline cascade: it waits for the VAD endpoint of each segment, identifies the segment’s language and then runs Qwen3-ASR-1.7B and Qwen3.8-27B with the offline prompt, so its EndOffset includes the endpoint silence and the compute time. In Table 6, the same-backbone cascade feeds the transcripts of M1, MODEL’s stage-2 checkpoint (Section 5), to the LocalAgreement-2 translation stage, and the size-matched cascade pairs Qwen3-ASR-1.7B with an LLM of about 1.7B parameters, the size of MODEL’s language model. The diarization cascades of Table 5 combine pyannote (Bredin, 2023) with offline Qwen3-ASR-1.7B, and Streaming Sortformer (Medennikov et al., 2025) with streaming Nemotron 3.5 ASR. *[Draft: planned]*

Operating points. Each system with a latency knob runs at two operating points, chosen once on the CST-Bench-Syn calibration split (about 5% of its sessions) by a rule that is fixed before any test run and is the same for all systems. We sweep the knob under the oracle wrapper, so that one setting serves both wrappers, and for each of two budgets $\Lambda_{\text{low}} < \Lambda_{\text{high}}$ on the mean StreamLAAL ([TBD]) take the setting with the highest COMET (chrF for Korean targets) that stays within the budget; a system that cannot meet a budget runs at its fastest setting, which we mark. The consecutive cascade, which exists only in realistic form, sweeps the minimum silence of its own VAD instead, and MODEL’s two operating points are values of δ_t chosen by the same rule. The offline systems and the streaming bound have a single operating point. *[Draft: planned; until CST-Bench-Syn is built, systems run at their documented default settings and are re-run once the calibration split exists]*

Release. We release the wrapper code with the scorer, including segmentation, language identification, routing and the settings of every operating point, so that any single-direction system can be evaluated under the same protocol. For CST-Bench-TAXI, we release per-turn scores keyed by turn ID but no system outputs (Appendix H).

E. Additional Results

This appendix gives the full numbers behind Figures 5 and 9 (Table 13), a dose-response analysis over the inter-turn gap (Figure 9), latency by source-turn duration (Figure 10), computation-aware latency (Table 14) and a failure taxonomy (Table 15). All analyses follow the protocol of Section 4: 95% session-clustered bootstrap intervals, and no threshold

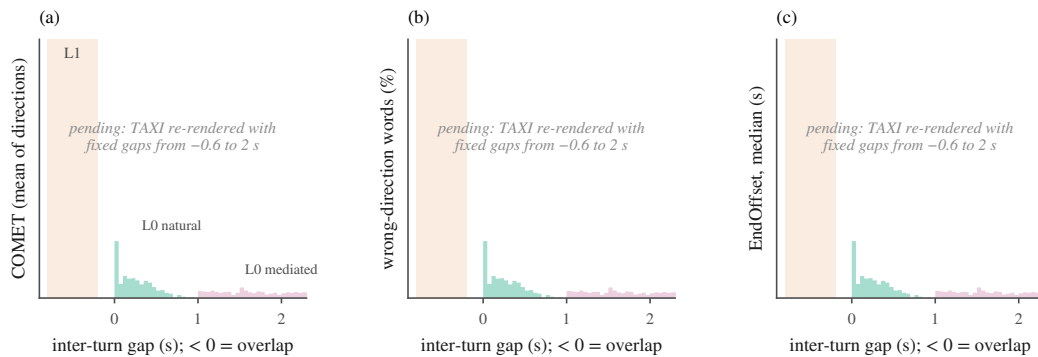


Figure 9. Gap dose-response: CST-Bench-TAXI re-rendered with one fixed gap at every speaker change, from -0.6 s (overlap) to 2 s. (a) COMET (mean of the two directions), (b) wrong-direction words and (c) median EndOffset for the best pipeline (solid: oracle segmentation; dashed: VAD and language-ID routing) and MODEL. Background histograms show the gaps drawn at the 540 speaker changes under L0 natural (green) and L0 mediated (pink); the shaded band is the overlap range of L1, whose next turn starts 0.2 – 0.8 s before the previous one ends. Numbers in part (c) of Table 13. [Draft: results pending; the fixed-gap renders are planned]

or operating point tuned on TAXI. So far, only the offline rows on CST-Bench-TAXI L0 natural (preliminary) and a few timings exist. [Draft: every other cell is pending; L1, the fixed-gap renders and CST-Bench-Syn are planned]

Full grid. For every timing condition, language pair and direction, Table 13 lists the systems run under it, each at its pre-registered main operating point; for CST-Bench-TAXI L0 natural, part (a) also gives the intervals of the corresponding rows of Table 3. The offline systems receive gold turns and translate each turn after it ends. Their StreamLAAL therefore equals the mean source-turn duration (4.624 s for En $\rightarrow$ De and 3.086 s for De $\rightarrow$ En on CST-Bench-TAXI), and their EndOffset is 0 on the ideal clock because their output is time-stamped at the gold turn end; Table 14 adds their compute. The gold-transcript row isolates translation quality, the Qwen3-ASR row adds the recognition of telephone-band speech, and Whisper ST covers De $\rightarrow$ En only. Under L1, an oracle segment cut from the mono mix also contains the overlapping speech of the other speaker; the oracle-separation row of Table 4 removes it. On CST-Bench-Syn we interpret only rankings and effects relative to each pair’s own offline bound, never absolute scores across pairs, because one side of every pair is translationese.

Gap dose-response. Each timing condition draws its gaps from one distribution. To test whether pipelines degrade smoothly as turns follow each other more closely, we re-render CST-Bench-TAXI with one fixed gap g at every speaker change, $g \in \{-0.6, -0.4, -0.2, 0.05, 0.2, 0.5, 1.0, 2.0\}$ s, each rendering under its own configuration name and checksums; a negative gap starts the next turn $|g|$ before the previous one ends, as in L1. We run the best pipeline of Table 4 with both wrappers, and MODEL (Figure 9; numbers in part (c) of Table 13). A smooth trend would show that the conclusions do not hinge on the chosen L0 and L1 distributions; a sharp drop would locate the gap at which VAD-based routing breaks down. From $g = 2$ s to $g = -0.6$ s, COMET changes by [TBD] with the realistic wrapper, by [TBD] with oracle segmentation and by [TBD] for MODEL.

Why simultaneous? Latency by turn length. With oracle endpoints, offline systems reach an EndOffset of 0 on the ideal clock, which raises the question of why simultaneity is needed at all. A deployed consecutive system must first detect that a turn has ended, so the listener waits for the endpoint detector and, on the computation-aware clock, also for the translation of the whole turn. Lag within a turn grows with the turn as well: a system that emits a turn’s translation only after the turn ends scores a StreamLAAL of at least the turn’s duration (Section 4). CST-Bench-TAXI turns are short (median 2.86 s), but 22.5% of them exceed 5 s (Table 7). Figure 10 therefore breaks median EndOffset and StreamLAAL down by source-turn duration for the realistic consecutive cascade, the best realistic streaming pipeline and MODEL. For turns longer than 6 s, the median EndOffset is [TBD] s for the consecutive cascade, [TBD] s for the best streaming pipeline and [TBD] s for MODEL.

Computation-aware latency. Main-text latencies use the ideal clock. Table 14 times every system on one H200 GPU, as a single stream without batching, and re-scores it on the computation-aware clock, on which an emission time also counts the processing time spent until the emission (Ma et al., 2020c). Streaming systems receive the audio in 80 ms chunks; we report the median and the 99th percentile of the processing time per chunk (tick) and the real-time factor (RTF).

Table 13. Full results by timing condition and language pair: the numbers behind Figures 5 and 9 (ideal clock; each system at its pre-registered main operating point). Part (a) covers CST-Bench-TAXI; parts (b) and (c) follow. Seg.: oracle = gold turns and source language; realistic = VAD and language-ID routing; none = the system segments the stream itself. Offline: gold→LLM and ASR→LLM translate the gold or the Qwen3-ASR-1.7B transcript of each turn with Qwen3.8-27B, and Whisper ST is Whisper-large-v3 translating into English; streaming cascade: streaming ASR and Qwen3.8-27B with LocalAgreement-2; M4T v2: SeamlessM4T v2 (Table 12). X is German on CST-Bench-TAXI and CST-Bench-Syn En↔De, and Korean on CST-Bench-Syn Ko↔En. Latencies are in seconds over non-empty turns: mean StreamLAAL and median EndOffset per direction; wrong-direction words, median switch latency and empty turns are pooled over both directions; Table 14 lists each system’s encoder look-ahead ℓ_{enc} . Each estimate will carry the half-width of its 95% session-clustered bootstrap CI as a subscript. Offline systems translate each oracle turn after it ends, so their StreamLAAL equals the mean source-turn duration and their EndOffset is 0 by construction. †preliminary; to be re-run under the final protocol (same runs as Table 3; no CIs yet). -: not applicable; blank: pending. [Draft: L1 is not rendered yet]

System	Seg.	En→X				X→En				Both directions		
		COMET	chrF	Stream-LAAL	End-Offset	COMET	chrF	Stream-LAAL	End-Offset	Wrong dir. (%)	Switch latency	Empty turns (%)
<i>(a) CST-Bench-TAXI, L0 natural</i>												
Offline: gold→LLM	oracle		71.72 [†]	4.62 [†]	0		58.05 [†]	3.09 [†]	0			
Offline: ASR→LLM	oracle		67.97 [†]	4.62 [†]	0		56.61 [†]	3.09 [†]	0			
Offline: Whisper ST	oracle	—	—	—	—		56.67 [†]	3.09 [†]	0			
Consecutive cascade	realistic											
SeamlessStreaming	oracle											
SeamlessStreaming	realistic											
Streaming cascade	oracle											
Streaming cascade	realistic											
M4T v2 + AlignAtt	oracle											
M4T v2 + AlignAtt	realistic											
MODEL	none											
<i>CST-Bench-TAXI, L0 mediated</i>												
Offline: gold→LLM	oracle											
Offline: ASR→LLM	oracle											
Offline: Whisper ST	oracle	—	—	—	—							
Consecutive cascade	realistic											
SeamlessStreaming	oracle											
SeamlessStreaming	realistic											
Streaming cascade	oracle											
Streaming cascade	realistic											
M4T v2 + AlignAtt	oracle											
M4T v2 + AlignAtt	realistic											
MODEL	none											
<i>CST-Bench-TAXI, L1 (turn-end overlap)</i>												
Offline: gold→LLM	oracle											
Offline: ASR→LLM	oracle											
Offline: Whisper ST	oracle	—	—	—	—							
Consecutive cascade	realistic											
SeamlessStreaming	oracle											
SeamlessStreaming	realistic											
Streaming cascade	oracle											
Streaming cascade	realistic											
M4T v2 + AlignAtt	oracle											
M4T v2 + AlignAtt	realistic											
MODEL	none											

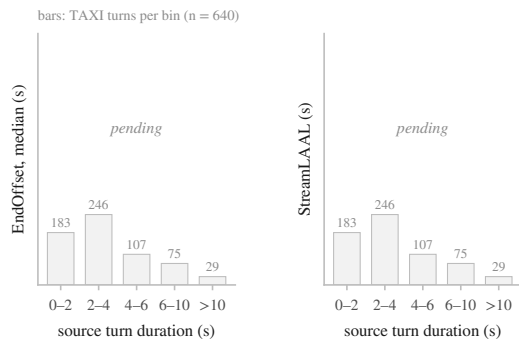


Figure 10. Why simultaneous? Median EndOffset (left) and StreamLAAL (right) by source-turn duration on CST-Bench-TAXI L0 natural, for the realistic consecutive cascade, the best realistic streaming pipeline and MODEL. Bars: CST-Bench-TAXI turns per duration bin (183, 246, 107, 75 and 29 of 640). [Draft: results pending]

Table 14. Computation-aware latency and real-time factor (RTF) on one H200 GPU, single stream, no batching; CST-Bench-TAXI L0 natural, pooled over both directions. ℓ_{enc} : encoder look-ahead, counted at its bound. Tick: processing time per 80 ms input chunk; RTF: processing time over audio duration. A real-time claim needs a measured RTF below 1; a tick above 80 ms means that the output falls behind the input for a while. Ideal clock: an emission time is the audio consumed; computation-aware (CA) clock: it also counts the processing time spent until the emission (Ma et al., 2020c). StreamLAAL is a mean and EndOffset a median over non-empty turns. The batched offline row comes from the preliminary offline run, with approximate compute; the context rows are earlier measurements of the single-speaker backbone with the 0.6B thinker (not a CST system), taken outside this protocol. [†]preliminary; to be re-run under the final protocol. —: not applicable or not reported; blank: pending. [Draft: planned except the batched offline row and the context rows]

System	Hardware	ℓ_{enc} (ms)	Tick (ms)		RTF	StreamLAAL (s)		EndOffset (s)	
			p50	p99		ideal	CA	ideal	CA
MODEL (1.7B thinker)	H200	80							
SeamlessStreaming	H200								
Streaming cascade (LA-2)	H200								
SeamlessM4T v2 + AlignAtt	H200								
Consecutive cascade (unbatched)	H200	—	—	—					
Offline: ASR→LLM, per turn (unbatched)	H200	—	—	—				0	
Offline: gold/ASR→LLM, batched	—	—	—	—	—	—	—	0	0.1–0.2 [†]
Context: single-speaker ASR backbone, 0.6B thinker (not a CST system)									
0.6B backbone	H200	80	≈60	100–150	—	—	—	—	—
0.6B backbone, MLX fp16 thinker	M4 Mac	80	56	—	0.82	—	—	—	—

Consecutive and offline systems are timed per turn. The only timings so far were taken outside this protocol. The single-speaker backbone with the 0.6B thinker takes about 60 ms per chunk at the median on an H200 but 100–150 ms at the 99th percentile, so it misses the 80 ms budget there; on an M4 Mac with the thinker in MLX (fp16), it takes 56 ms per chunk at the median and runs at an RTF of 0.82. In the preliminary offline run, batched compute moved the median EndOffset of the offline systems from 0 to at most about 0.2[†] s ([†]preliminary; to be re-run under the final protocol). The 1.7B thinker has not been timed, and we claim real time only for a system with a measured RTF below 1.

Failure taxonomy. Table 15 counts six failure types per 100 source turns. The detectors work on the scorer’s alignments and timelines, so the table reports counts without showing any TAXI text, and their thresholds are fixed before any system is scored. Merged turns, in which one VAD segment or output piece spans turns of both speakers, send one speaker’s words to the wrong channel or delay the switch. A truncated tail leaves the last words of a turn untranslated, for example when a segment is closed early. Hallucinations are output words without an aligned reference word and with low reference-free quality (Rei et al., 2022b), cross-checked against the critical-error spans of xCOMET (Guerreiro et al., 2024). We estimate the precision of each detector by hand on 50 CST-Bench-Syn turns per condition, and any example we show comes from CST-Bench-Syn. Under L1, the best realistic pipeline merges [TBD] turns per 100 (L0 natural: [TBD]), and MODEL merges [TBD].

Table 15. Failure taxonomy: counts per 100 source turns on CST-Bench-TAXI L0 natural (L0) and L1, found by automatic detectors on the scorer’s alignments; pipelines run with the realistic wrapper. Each detector’s precision is checked by hand on 50 CST-Bench-Syn turns per condition, never on TAXI text. Consec.: consecutive cascade; Stream. casc.: streaming cascade (LA-2); M4T+AA: SeamlessM4T v2 + AlignAtt. Thresholds are fixed before any system is scored. If the optional L1b stress test is adopted, a row for translated backchannels is added. Blank: pending. [Draft: planned]

Failure type	Detector	Consec.		Seamless		Stream. casc.		M4T+AA		MODEL	
		L0	L1	L0	L1	L0	L1	L0	L1	L0	L1
Merged turns	One VAD segment, or one output piece, spans turns of both speakers										
Wrong direction	Words on the wrong channel for the active turn, in the wrong language, or copied from the source (scorer detector)										
Truncated tail	The final words of the turn’s reference stay unaligned										
Empty turn	No output word is assigned to the turn										
Hallucination	Output with no aligned reference word and low reference-free QE (Rei et al., 2022b); xCOMET critical-error spans (Guerreiro et al., 2024) as a cross-check										
Late switch	Switch latency above 2 s										

Second-best pipeline and diarization without a collar. Repeating the factorial of Table 4 for the second-best pipeline ([TBD]) changes its COMET under L1 by [TBD] when oracle turns replace VAD and by [TBD] when the oracle language replaces language identification. Table 5 scores DER with a 0.25 s collar. Without a collar, every boundary error counts at the 80 ms resolution; DER on CST-Bench-TAXI L0 natural is then [TBD] for MODEL with $\delta_s = 2$ and [TBD] for the streaming diarization cascade, and [TBD] and [TBD] under L1.

F. Model and Backbone Details

This appendix specifies the streaming backbone of MODEL, the only part that is built, and evaluates it on single-speaker English and Korean transcription; none of its numbers is a CST result. Table 16 lists the training stages, Table 17 and Figure 12 report the backbone, and Figure 11 works through the delay rule of Section 5 on a recorded utterance.

Encoder. The encoder of Nemotron 3.5 ASR Streaming 0.6B (NVIDIA, 2026) is a cache-aware FastConformer (Gulati et al., 2020; Rekesh et al., 2023; Noroozi et al., 2024): 128-bin log-mel features (25 ms window, 10 ms hop), depthwise-striding subsampling by 8 and 24 layers with $d_{\text{model}} = 1024$ and 8 heads, so one frame covers 80 ms and a stream of T s yields $K = \text{round}(12.5T)$ frames. Its attention context $[56, N]$ gives the number of 80 ms frames visible to the left and to the right. We use $[56, 0]$, i.e. 4.48 s of left context and none on the right; the released checkpoint defaults to $[56, 3]$, which adds 240 ms of right context. With $[56, 0]$, encoding a whole utterance at once yields the same frames as streaming, so training encodes each batch without a cache, and live inference uses the cache-aware step, whose output is frame-identical. The encoder’s own RNN-T decoder (Graves, 2012) serves only as a baseline (Table 17). When the encoder is trained, its attention context is sampled per step from $[56, 0]$, $[56, 3]$, $[56, 6]$ and $[56, 13]$, so one checkpoint can be evaluated at several look-aheads (Figure 12).

Effective look-ahead. A nominal context can hide look-ahead, so we measure it. We encode a 14.84 s LibriSpeech utterance from its full features and from features truncated at 3, 6, 9 and 12 s and compare the outputs frame by frame (fp32, TF32 disabled, relative tolerance 10^{-3}); the effective look-ahead is the span before the cut in which the outputs differ. With $[56, 0]$, only the last frame changes, through the boundary of the subsampling convolution, so the look-ahead is 80 ms at every cut; with $[56, 3]$, it is 160, 240 and 320 ms (minimum, mean and maximum over the cuts). Two strictly causal reference models give 0 ms, which validates the test. The test also shows why Qwen3-ASR’s own audio encoder is removed: in the implementation we tested, its attention spans the whole utterance, and even its intended mode with 1 s blocks looks ahead by 420 ms on average and by up to 800 ms.

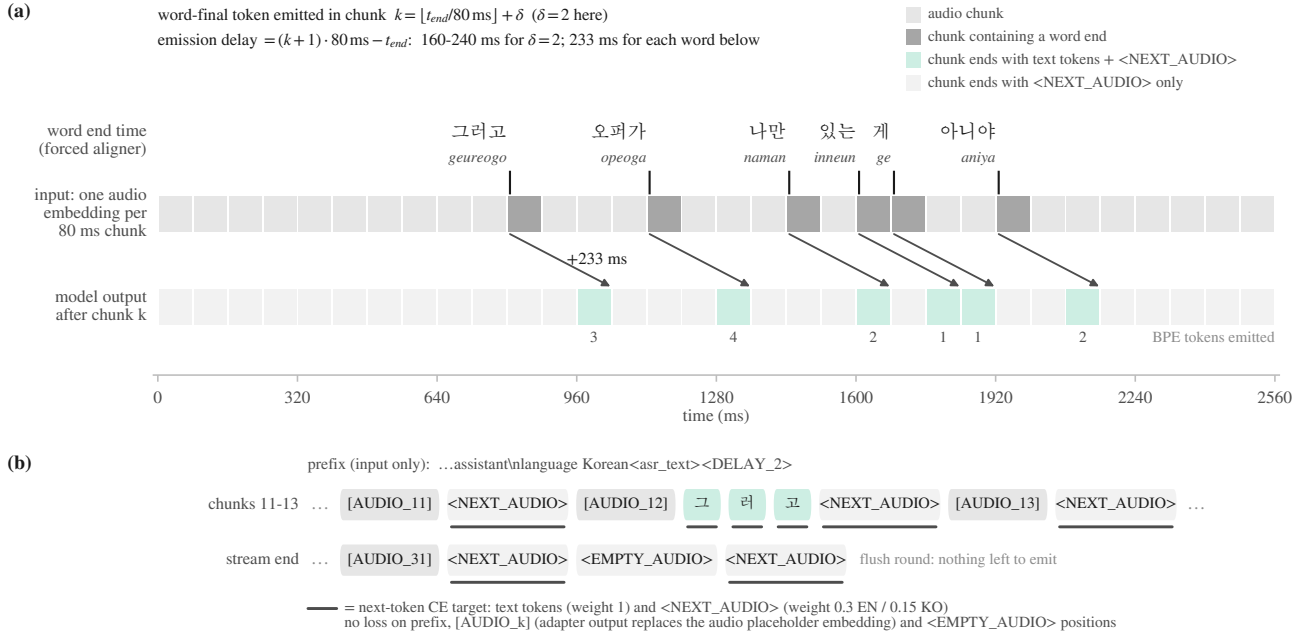


Figure 11. Delay rule of the single-speaker backbone on a recorded KsponSpeech utterance (2.6 s, $K = 32$ chunks, $\delta = 2$). (a) One audio embedding enters per 80 ms chunk. A word that ends at t_{end} (forced alignment) is emitted, with all its BPE tokens, after chunk $k = \lfloor t_{\text{end}}/80 \text{ ms} \rfloor + \delta$, so its emission delay $(k + 1) \cdot 80 \text{ ms} - t_{\text{end}}$ lies between 160 and 240 ms for $\delta = 2$; here it is 233 ms for all six words. Numbers under the output row count the BPE tokens emitted. (b) Serialized LM sequence for chunks 11–13 and the end of the stream. Underlined tokens are next-token targets, weighted 1 for text and 0.3 (English) or 0.15 (Korean) for the end-of-chunk token (<next>) (NEXT_AUDIO); the prefix, the audio positions and <empty> (EMPTY_AUDIO) carry no loss.

Adapter and thinker. The adapter is LayerNorm, Linear (1024 $\rightarrow$ 2048), GELU and Linear (2048 $\rightarrow$ h), where h is the hidden size of the thinker: 1024 for the 0.6B thinker and 2048 for the 1.7B one (4.2M and 6.3M adapter parameters). The 0.6B thinker is a Qwen3 decoder with 28 layers, 16 query and 8 key-value heads of dimension 128, a feed-forward size of 3072, a vocabulary of 151,936 tokens and tied input and output embeddings. The adapter output replaces the input embedding at each audio placeholder position (token 151676), one per 80 ms chunk. With the 1.7B thinker, Stage 1 trains 2,336M parameters: thinker, encoder and adapter.

Sequence and special tokens. The input-only prefix follows the chat format of Qwen3-ASR with an empty system turn; the assistant turn opens with the line “language English” or “language Korean”, the transcript marker and <DELAY_ δ >. The prefixes of the two languages differ only in that line, so the language is an input and the backbone performs no language identification. Each chunk k then contributes a_k , the text tokens due in it and the end-of-chunk token <next> (NEXT_AUDIO in Figure 11). After the last chunk, a flush round (<empty>, the remaining text tokens, <next>) runs if tokens remain, and a final round (<empty>, <next>) always runs; <empty> (EMPTY_AUDIO) is an input placeholder that carries no loss. Twelve special tokens use spare rows of the embedding matrix, so the vocabulary is not resized, and each is initialised as the mean embedding plus $0.02\mathcal{N}(0, 1)$ noise: <next> (id 151705), <empty> (151706), the speaker tags <A> and (151707 and 151708), which are reserved but untrained and blocked at decoding, and <DELAY_1> to <DELAY_8> (151709–151716). In the 0.6B model, token ids end at 151716 while the embedding matrix has 151,936 rows, which leaves 219 unused rows for the target-language tags and <TDELAY_ δ_t > of Section 5. [Draft: planned: these tags are not yet allocated]

Targets, loss and decoding. Targets are normalised lexical transcripts (lowercase, no punctuation, numbers spelled out). Word end times come from Qwen3-ForcedAligner (Shi et al., 2026) at 80 ms resolution, and all tokens of a word are due in the chunk that the delay rule assigns to the word’s end; tokens due at or after the last chunk go to the flush round, and tokens due in the same chunk keep their end-time order. Training imposes no limit on the tokens per chunk. On LibriSpeech and KsponSpeech, targets average 0.22 and 0.27 text tokens per chunk, and 75–80% of the chunks emit only <next>, hence its lower loss weight. The output layer is computed only at labelled positions, which bounds memory on long sequences. Decoding is greedy with a KV cache; it forces <next> after 8 tokens in a chunk or $6K + 64$ tokens in a stream, runs flush

Table 16. Training stages, data and hyperparameters. Stages 0 and 1 exist, and the one-epoch Stage-1 pilot is the backbone of Table 17; Stages 2 and 3 are planned (M1–M3 as in Section 5), and their open settings are blank. Hours are approximate; data sources and augmentation are detailed in Appendix F. lr: peak learning rate after a linear warm-up.

Stage	Trainable	Data	Targets	Schedule	Hardware	Status
0: adapter warm-up	adapter (6.3M); thinker frozen	English and Korean ASR	final mode only (Qwen3-ASR format)	2,000 steps, warm-up 100; 16,384-token batches	1 node	built
1: streaming and final-mode ASR	encoder, adapter, thinker (2,336M)	≈ 3.8 k h curated English and Korean speech + ≈ 4.3 k h English and Korean ASR corpora (806 h English); no German; augmented	lexical transcripts; $\delta \in \{2, 3, 4, 6, 8\}$; final mode with probability 0.3	1 epoch = 3,980 steps; lr 6×10^{-5} thinker (warm-up 500), 10^{-3} adapter, 10^{-5} encoder	2 nodes $\times$ 8 GPUs	built (pilot)
2 (M1): two-speaker bilingual ASR		+ German ASR, 8 kHz telephone-band augmentation, simulated two-speaker En $\leftrightarrow$ De and En $\leftrightarrow$ Ko mono sessions	speaker-tagged transcripts at delay δ_s ; two-slot speaker activity			planned
3 (M2, M3): translation streams		+ CoVoST 2 and MuST-C (En $\leftrightarrow$ De); silver teacher translations in four directions	translations committed at turn ends (M2) or at MU2 (M3), delay δ_t			planned

rounds until one is empty (at most 8), and blocks the chat-format, audio, transcript-marker, $\langle \text{empty} \rangle$, speaker and delay tokens. In the trained backbones, $\delta = 2/3/4$ gives a median emission delay of about 200/280/360 ms, and at most 0.5% of English emissions (2–5% of Korean ones) come 80 ms or more before their target chunk. Figure 11 shows the rule on a 2.6 s utterance: its six words end at 807, 1127, 1447, 1607, 1687 and 1927 ms and are emitted after chunks 12, 16, 20, 22, 23 and 26, each 233 ms after its end; chunks 0–11 and 27–31 emit only $\langle \text{next} \rangle$, and the flush round is empty.

Training. Stage 0 trains only the adapter, with the thinker frozen, in final mode (the native Qwen3-ASR format), because a randomly initialised adapter otherwise causes very large gradients in the first steps. Stage 1 starts from Stage 0 and the original Qwen3-ASR thinker and trains all parts for one epoch, with $\delta \in \{2, 3, 4, 6, 8\}$ and final mode for each utterance with probability 0.3 (Table 16). Its data are about 3.8 k h drawn from a quality-controlled pool of 19 English and Korean sources, plus about 4.3 k h of ASR corpora: 806 h of English from LibriSpeech, VoxPopuli, Granary-YODAS (Koluguri et al., 2025) and Switchboard (Godfrey et al., 1992), and about 3.5 k h of Korean from KsponSpeech, NIKL conversational speech and AI Hub conversational and broadcast speech; there is no German. We augment with SpecAugment (2 frequency masks of up to 27 bins and 2 time masks of up to 3% of the utterance), speed perturbation (0.9, 1.0, 1.1), noise (probability 0.4, SNR 5–30 dB), music (SNR 10–30 dB) and reverberation (probability 0.2). The noise bank holds 8.9 h of noise (MUSAN, DEMAND, ETSI), 42.6 h of music (MUSAN) and 6,325 simulated and real room impulse responses; reverberation is applied before noise, and the direct-path peak of each impulse response is aligned to $t = 0$ so that word times stay valid. Stronger time masking (about 12% of the frames) lowered the top-1 accuracy on text tokens at the first step from 0.985 to 0.84, so we avoid it. One epoch takes 3,980 steps on 2 nodes with 8 GPUs each, at about 2.5 s per step including I/O stalls (0.6–1.0 s of compute); the training loss ended at 0.27 and was still falling. The 0.6B pilot uses the same data, steps and augmentation. After training, we reload each checkpoint and compare it with the model in memory key by key, encoder included; for both pilots no key differs, including all 638 encoder tensors. *[Draft: longer Stage-1 runs exist but stay out of the paper until their numbers are recorded with a per-epoch analysis]*

Planned stages. Stage 2 (M1) starts from the 1.7B Stage-1 pilot. Before training it, we fixed three acceptance criteria: single-speaker WER and CER at most 5% (relative) above those of the Stage-1 pilot on VoxPopuli-AA and KsponSpeech, cpWER on synthetic L0 sessions at most 1.2 times the matched single-speaker WER, and a speaker-attribution error of at most 5%. Stage 3 then adds the translation streams (Section 5). *[Draft: planned; German data still has to be obtained, and if M1 misses the criteria twice, M1 transcripts translated by an LLM become our system]*

Evaluation protocol. Backbone results use a single-turn protocol: each utterance is followed by 1 s of digital silence, the state is reset per utterance, and decoding is greedy in fp32 with TF32 disabled. English is scored by WER and Korean

Table 17. Streaming ASR backbone on single-speaker English and Korean speech (not CST): error rates in % at decoder delay $\delta = 4$ and $\delta = 6$ and in final mode, encoder context [56, 0], single-turn protocol. WER on English VoxPopuli-AA, together with AA-WER, our approximation of the scoring rules published with this test set; CER without spaces on KsponSpeech eval_clean and eval_other. *Backbone of MODEL. The two Stage-1 models share data, steps and augmentation (Table 16); the earlier 0.6B models are described in Appendix F. Qwen3-ASR-1.7B is non-causal and transcribes whole utterances; the RNN-T is our encoder with its own causal transducer decoder. Both have one operating point. VoxPopuli-AA is not speaker-disjoint from our training data: 80 of its 207 speakers and 280 of its 292 sessions also occur in training. Paired-bootstrap comparisons are given in the text. —: no final mode.

Model	VoxPopuli-AA WER			VoxPopuli-AA AA-WER			Kspon clean CER			Kspon other CER		
	$\delta = 4$	$\delta = 6$	final	$\delta = 4$	$\delta = 6$	final	$\delta = 4$	$\delta = 6$	final	$\delta = 4$	$\delta = 6$	final
Stage 1, 1.7B thinker*	7.70	7.18	6.84	5.73	5.20	4.77	9.70	9.40	8.21	9.95	9.63	8.42
Stage 1, 0.6B thinker	7.73	7.18	7.14	5.73	5.09	5.09	10.55	10.24	9.27	10.87	10.47	9.22
Earlier recipe, 0.6B	8.94	8.01	—	7.18	6.16	—	12.32	11.76	—	12.86	12.67	—
Earlier + commit tokens, 0.6B	9.88	9.15	—	8.02	7.26	—	12.26	12.04	—	12.26	12.19	—
Qwen3-ASR-1.7B, offline	5.10			3.22			8.92			8.71		
Nemotron RNN-T, [56, 0]	6.88			4.88			19.46			16.80		

by CER without spaces, after normalising the text with numbers spelled out. VoxPopuli-AA is a cleaned release of the VoxPopuli test set (628 utterances, 1.98 h, 292 sessions, 18,072 reference words); AA-WER approximates the scoring published with it, part of which is not public: we apply Whisper’s text normalisation (Radford et al., 2023), split digits and weight the mean by utterance duration. KsponSpeech eval_clean and eval_other have 3,000 and 2,687 utterances. Paired bootstrap tests use 2,000 resamples, clustered by session on VoxPopuli-AA and by utterance on KsponSpeech; we call a difference significant if its 95% interval excludes zero. Differences and relative changes are computed from unrounded error rates.

Backbone results. Table 17 compares the two Stage-1 pilots with two earlier 0.6B models and two references. The earlier recipe trained a 0.6B model in several stages on 5.6k h of English and Korean ASR data with $\delta \in \{2, 3, 4, 6\}$ and without final mode; the second earlier model fine-tunes it with sentence-level commit tokens while the encoder is frozen. At $\delta = 4$ and $\delta = 6$, the 1.7B pilot improves on these two models by 10–24% and 19–22% (relative), all significant; only at $\delta = 2$ on KsponSpeech is the commit-token model better, by 0.7–1.0 points. Against the 0.6B pilot, the 1.7B thinker is 6–11% better in every Korean comparison, on par in English streaming, and 4% better in English final mode (6.84 vs. 7.14%; $\Delta = -0.31$ points, 95% CI $[-0.57, -0.03]$). Against the released Nemotron 3.5 model at [56, 0] (the encoder we start from, with its own causal RNN-T decoder), the English gap closes as δ grows: +4.87 points (+70.8%) at $\delta = 2$, +0.82 (+12.0%) at $\delta = 4$, +0.30 (n.s.) at $\delta = 6$ and -0.04 (n.s.) in final mode. On KsponSpeech, the backbone is 24.0% ($\delta = 2$) to 57.8% (final mode) better on eval_clean and 11.9% to 49.9% better on eval_other. In final mode, it beats offline Qwen3-ASR-1.7B on KsponSpeech eval_clean (8.21 vs. 8.92%; $\Delta = -0.71$, $[-1.05, -0.37]$), matches it on eval_other (8.42 vs. 8.71%, n.s.) and trails it in English by 1.73 points (6.84 vs. 5.10%, +33.9%).

Caveats. These are single-speaker results. VoxPopuli-AA is not speaker-disjoint from our training data, because the official VoxPopuli split is not: 80 of its 207 speakers and 280 of its 292 sessions also occur in training, although no utterance or sentence is duplicated. AA-WER only approximates the published scoring. The final-mode gain over offline Qwen3-ASR on KsponSpeech may be an in-domain effect, since the KsponSpeech training set is part of our data. The training data contain no German, and the backbone has not been evaluated on telephone-band CST-Bench-TAXI speech or on two-speaker mixtures, so none of these numbers carries over to CST.

Encoder look-ahead versus decoder delay. Latency can be bought with encoder right context or with decoder delay. As the 1.7B pilot was trained with sampled contexts, we evaluate the same checkpoint at [56, 0] and [56, 3] (Figure 12) and compare the two at their mean nominal delay: 80δ ms for [56, 0] and $80\delta + 120$ ms for [56, 3], whose explicit right context adds 120 ms on average and 240 ms at most. Below, Δ is the error of [56, 3] minus that of [56, 0] in points, given as VoxPopuli-AA WER / KsponSpeech eval_clean CER / eval_other CER, with significance from the paired bootstrap. Between 240 and 320 ms the evidence is mixed: [56, 3] at $\delta = 2$ (280 ms) is better in English than [56, 0] at $\delta = 3$ (240 ms) but worse in Korean ($\Delta = -0.49 / +0.51 / +0.58$, all significant), and worse everywhere than [56, 0] at $\delta = 4$ (320 ms; +0.77 / +1.49 / +1.65, all significant). Between 320 and 480 ms the two are interchangeable: [56, 3] at $\delta = 3$ (360 ms) differs from [56, 0] at $\delta = 4$ (320 ms) by $-0.18 / -0.21 / -0.14$ and from [56, 0] at $\delta = 6$ (480 ms) by $+0.34 / +0.09 / +0.17$,

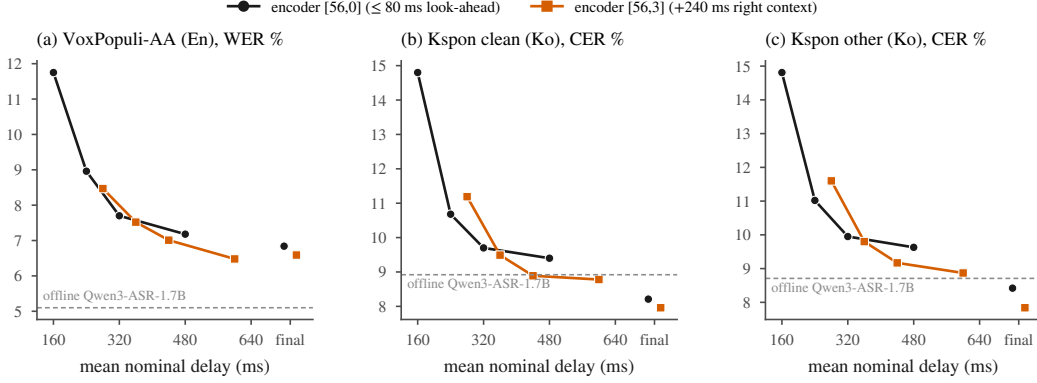


Figure 12. Encoder look-ahead versus decoder delay in the streaming backbone (Stage-1 pilot, 1.7B thinker): error rate against mean nominal delay with encoder context [56, 0] (look-ahead ≤ 80 ms; black circles) and [56, 3] (240 ms more right context; vermilion squares) for $\delta = 2, 3, 4, 6$, and in final mode; dashed: offline Qwen3-ASR-1.7B. (a) VoxPopuli-AA WER (English); (b) KsponSpeech eval_clean and (c) eval_other CER (Korean). The nominal delay is 80δ ms for [56, 0] and $80\delta + 120$ ms for [56, 3]. One checkpoint, trained with sampled encoder contexts, is evaluated at both settings. At matched mean delay, encoder look-ahead and decoder delay are largely interchangeable.

none significant. Right context helps from 440 ms on ([56, 3] at $\delta = 4$ vs. [56, 0] at $\delta = 6$: -0.17 (n.s.) / -0.51 / -0.46 ; at equal $\delta = 6$, i.e. 600 vs. 480 ms: -0.70 / -0.62 / -0.76 , significant) and in final mode (-0.25 / -0.25 / -0.58 , all significant), and matching the maximum rather than the mean delay would shrink these gains further. We therefore keep the causal [56, 0] and let δ set the latency, which MODEL conditions separately for each stream.

G. Translation-Unit Pilot

This appendix reports a text-level pilot that motivates the translation commit units of MODEL (Section 5). Its conditions differ from those of the benchmark: it simulates simultaneous translation over gold transcripts with gold word end times; it uses single-speaker Korean and English utterances from monolingual ASR corpora (150 per direction for quality, 300 per language for alignment statistics), not dialogues and not En $\leftrightarrow$ De; Qwen3.8-27B is the incremental translator; the references are offline translations by EXAONE-4.0-32B (silver references); and latency is computation-unaware LAAL on the gold word times. The pilot compares commit units under these conditions only and does not predict results for speech input, for trained models or on CST-Bench-TAXI.

Commit policies. Let $u_{1:n}$ be the gold transcript of an utterance, e_i the end time of word u_i , and $v_{1:J}$ the full offline translation by a teacher LLM. A policy commits translation text after it has read b source words, and committed text is never revised, unlike in re-translation (Arivazhagan et al., 2020). We compare four policies. (i) *Utterance end* commits once, after u_n , which amounts to offline translation of the utterance. (ii) *SEM.END* commits at sentence-level semantic-completion boundaries: the positions at which an existing LLM-based labeller, which sees only the source transcript, judges the transcript so far semantically complete, i.e., later words will not change its meaning (in practice mostly the sentence-final punctuation of the transcript); the utterance end is added. These boundaries do not depend on the target language. (iii) *Alignment-only meaning units (MU)*. Contextual alignment (Labiausse et al., 2025) assigns each target word v_j to the source position $\alpha_j = \arg \max_{1 \leq i \leq n} [L_j(i) - L_j(i-1)]$, where $L_j(i)$ is the summed log-probability of the tokens of v_j when $v_{1:J}$ is force-decoded from $u_{1:i}$; ties go to the earliest i . A position i is a *valid cut* if the target words aligned at or before i form a prefix $v_{1:k}$, so that, relative to the teacher’s translation, the output up to that point need not change later. An *MU boundary* is a valid cut that covers more target words than the previous valid cut; n is always a valid cut and an MU boundary. This approximates, with alignment instead of generation, the translation-prefix consistency that defines meaning units (“meaningful units” in Zhang et al., 2020): minimal source segments whose translation is a prefix of the full translation. The policy commits at every MU boundary. (iv) *Verified meaning units (MU2)* commit at a subset of the valid cuts. A candidate is a valid cut $b < n$ that covers more target words than the last commit b' and lies at least a minimum unit after it: 2 Korean eojeol (space-delimited words) or 3 English words, and $e_b - e_{b'} \geq 1.0$ s. At a candidate, the translator receives the source heard so far and the committed translation in its prompt, rather than as a forced prefix, and writes only the new text. A language guard regenerates once and then rejects output in the wrong script: Han or Kana characters,

Hangul in English output, or less than 30% Hangul in Korean output. A generation check, in the spirit of Zhang et al. (2020), accepts the new text only if its sentence-level chrF (Popović, 2015) against the newly covered span $v_{k'+1:k}$ of the teacher’s translation is at least $\theta = 50$, where k' is the coverage at the last commit. The rest of the utterance is translated at its end.

Setup. We sample 300 utterances per source language with word end times from monolingual Korean and English ASR corpora, plus a second, independent sample of the same size to check sample dependence; the source text is the gold transcript, and the SEM.END positions are the labeller’s output for it. The labeller has LLM judges rate candidate positions, proposed by an LLM or by rules (e.g., at sentence-final punctuation), and keeps a candidate if its ratings pass thresholds set per candidate type so that precision against LLM-annotated reference labels is at least 0.90 on a development split. Alignments come from Qwen3.8-27B on both samples and from EXAONE-4.0-32B on the first. The quality runs use 150 utterances per direction from the first sample (mean length 11.09 s for Korean and 15.05 s for English sources), with Qwen3.8-27B as translator and as the MU2 teacher. The references are EXAONE-4.0-32B offline translations of the full gold transcripts, from another model family than the translator; no human references exist for these utterances. Teachers and reference translator receive “*Translate the following $\langle src \rangle$ speech transcript into natural $\langle tgt \rangle$. Output only the $\langle tgt \rangle$ translation, nothing else.*”, followed by the transcript. [Draft: name the source corpora, the sampling arguments and the origin of the word times; the Korean transcripts seem to be filtered ASR output, so verify before calling them gold]

Block A of Table 18 runs the utterance end, SEM.END and MU2 with one incremental prompt and the language guard; only MU2 adds the minimum unit and the generation check. The prompt reads

*You are a simultaneous interpreter translating $\langle src \rangle$ speech into $\langle tgt \rangle$.
Source heard so far: $\langle source prefix \rangle$
Translation already given (cannot be changed): $\langle committed text, or “(none)” \rangle$
Translate only the part of the source that the translation already given does not cover yet, continuing it naturally in $\langle tgt \rangle$.
[The speaker is still talking: translate only what has been said so far and do not guess the rest.] Output only the new $\langle tgt \rangle$ text, nothing else. If nothing new can be translated yet, output nothing.*

where the bracketed sentence appears only before the end of the utterance, and a regeneration by the guard appends “*Write the answer in $\langle tgt \rangle$ only.*” Block B runs the utterance end, SEM.END and alignment-only MUs with the offline prompt, which before the end of the utterance also says “*The transcript may be incomplete because the speaker is still talking: translate only what has been said so far, do not guess or complete the rest.*”; the committed text is forced as the beginning of the answer, and the continuation is committed. Block B has no language guard, and its latency was not reported. In both blocks, the utterance-end translation may use up to 160 tokens and every other step at most $\min(160, 2\ell + 8)$, where ℓ is the token length of the utterance-end translation, and an m -gram ($m \leq 3$) repeated three or more times in a row is cut.

We report corpus BLEU (Papineni et al., 2002) and chrF with sacreBLEU (Post, 2018), with the BLEU tokenizer `intl` for Korean and `13a` for English output; chrF is primary for Korean output. An output *collapses* if it is empty, is in the wrong script as defined for the guard, or contains an m -gram ($m \leq 3$) repeated three or more times in a row. Latency ignores LLM computation and ASR delay. Output word j gets the commit time $d_j = e_b$ of the step that wrote it, and per utterance

$$\text{LAAL} = \frac{1}{\tau} \sum_{j=1}^{\tau} \left(d_j - (j-1) \frac{e_n}{|Y|} \right),$$

where $|Y|$ is the output length in words and τ the first j with $d_j = e_n$ (or $|Y|$ if there is none). This is the length-adaptive form (Papi et al., 2022) of Average Lagging (Ma et al., 2019), except that $|Y|$ replaces $\max(|Y|, |Y^*|)$ with the reference length $|Y^*|$; the two agree whenever the output is at least as long as the reference. *First commit* is d_1 . Both are averaged over utterances, so the LAAL of the utterance end equals the mean utterance length. [Draft: recompute LAAL with $\max(|Y|, |Y^*|)$ when the raw outputs are re-exported] For the alignment statistics, a source word u_i waits $e_{b(i)} - e_i$ until the first boundary $b(i) \geq i$ of a policy.

Verified meaning units halve the lag. In Block A (Table 18 and Figure 13), MU2 roughly halves LAAL relative to the utterance end, from 11.09 to 5.19 s (Ko→En) and from 15.05 to 7.33 s (En→Ko), stays below SEM.END (7.49 and 8.63 s), and makes its first commit earlier (7.03 against 11.09 s and 8.25 against 15.05 s). chrF falls from 61.3 to 60.2 and from 44.2 to 42.0; BLEU falls more, from 35.2 to 31.1 and from 23.4 to 19.9. Whether this reflects paraphrase or errors is untested, since the references are silver and no human judged the outputs. SEM.END scores close to the utterance end, but its output is identical to the utterance-end output in 96/150 (Ko→En) and 83/150 (En→Ko) utterances.

Table 18. Text-level commit-unit pilot (Appendix G). Blocks A and B: Ko↔En, 150 single-speaker utterances per direction, gold transcripts and word end times, Qwen3.8-27B as translator, silver references (offline EXAONE-4.0-32B translations) and computation-unaware LAAL. The two blocks use different prompts, so rows are compared only within a block. Block C (planned): En↔De CST-Bench-TAXI turns with gold transcripts and forced-aligned word times, scored by `cstbench eval` against the human references; its left group is De→En, its right group En→De, and its LAAL is StreamLAAL on the ideal clock. LAAL and First (commit time of the first output word) in seconds. Coll.: collapsed outputs (empty, wrong script or a repetition loop), out of 150 in Blocks A and B. Candidates accepted: MU2 candidates that pass the language guard and the chrF check. -: not reported; blank: pending.

Block / commit policy	Ko→En (C: De→En)					En→Ko (C: En→De)				
	BLEU	chrF	LAAL	First	Coll.	BLEU	chrF	LAAL	First	Coll.
<i>A: prompt-based commits</i>										
Utterance end	35.2	61.3	11.09	11.09	0	23.4	44.2	15.05	15.05	1
SEM.END	35.3	61.1	7.49	8.23	0	22.4	43.7	8.63	9.74	0
Verified MU (MU2)	31.1	60.2	5.19	7.03	0	19.9	42.0	7.33	8.25	1
candidates accepted	197/368 (54%)					185/873 (21%)				
<i>B: forced-prefix continuation (latency not reported)</i>										
Utterance end	32.9	58.4	–	–	–	21.2	42.3	–	–	–
SEM.END	34.4	59.2	–	–	–	19.2	40.8	–	–	–
Alignment-only MU	28.1	54.2	–	–	5	12.9	33.1	–	–	31
<i>C: En↔De CST-Bench-TAXI turns, human references (planned)</i>										
Turn end										
Verified MU (MU2)										
candidates accepted										

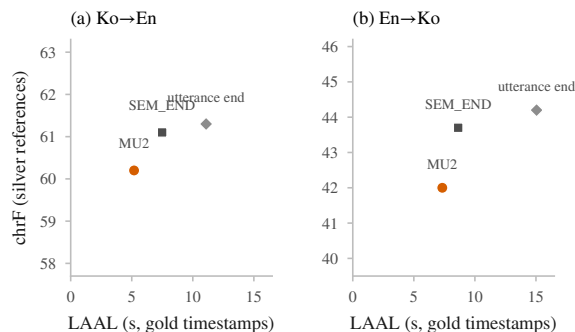


Figure 13. Commit-unit pilot, Block A of Table 18: chrF against computation-unaware LAAL for (a) Ko→En and (b) En→Ko (text level; 150 utterances per direction; gold transcripts and word times; silver references). Verified meaning units (MU2) roughly halve LAAL relative to committing at the utterance end, at a small chrF cost; SEM.END lies in between on both axes. The LAAL of the utterance end equals the mean utterance length.

SEM.END is sentence-level. Across the three alignment runs, an utterance contains 1.16–1.44 SEM.END units against 3.09–3.45 MUs for Korean sources, and 1.75–1.88 against 5.52–5.73 for English sources. A SEM.END unit spans 11.83–12.82 eojeol against 5.70–6.73 for an MU in Korean, and 24.68–26.15 against 9.22–9.93 words in English (Figure 14). Of all SEM.END positions, 42.3–85.9% coincide with the utterance end, which is a valid cut by definition; counting these, 79.6–92.2% are valid cuts. Yet SEM.END positions cover only 2.0–8.1% of the MU boundaries inside utterances, and a source word waits 3.75–4.05 s (Korean) and 5.20–5.73 s (English) for the next SEM.END boundary, against 2.59–3.25 s and 3.22–3.38 s for the next MU boundary. Semantic completion of the source therefore marks few of the points at which a translation could already be committed. [Draft: also report the valid-cut share of the SEM.END positions inside utterances]

Unverified units collapse, and acceptance depends on the direction. Block B commits without verification: alignment-only MUs continued from a forced prefix collapse in 5/150 Ko→En and 31/150 En→Ko outputs, through text in another script (including copied source), loops or empty output, and their chrF drops to 54.2 and 33.1, against 58.4 and 42.3 for the utterance end of the same block. MU2 collapses in 0/150 and 1/150 utterances; the single En→Ko case is an utterance whose utterance-end output is empty as well. MU2 differs from Block B’s MUs in its prompt, the guard, the minimum unit and the check at once, so the pilot does not separate their effects. The check accepts 197 of 368 candidates (54%) for Ko→En but only 185 of 873 (21%) for En→Ko; nearly all rejections fail the chrF threshold (171 and 685; the other

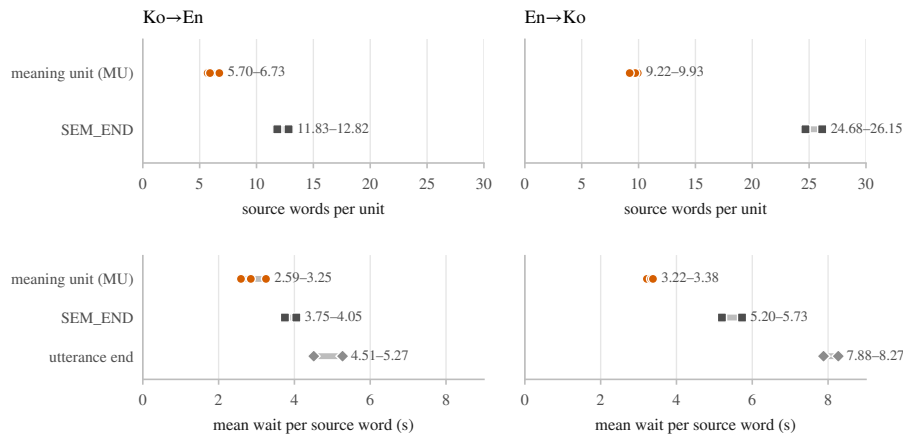


Figure 14. Unit granularity on 300 utterances per source language: source words per unit (Korean: eojeol) and mean wait of a source word until the next commit boundary, for alignment-only meaning units (MU), SEM_END and the utterance end. Dots are three alignment runs (Qwen3.8-27B and EXAONE-4.0-32B on one sample; Qwen3.8-27B on a second sample); bars span the minimum to the maximum. SEM_END and utterance-end values depend on the sample but not on the teacher.

three En→Ko rejections are one empty and two wrong-language outputs). This is consistent with Korean word order: a Korean translation places the verb last, so the translation of an English prefix rarely matches the span of the full translation aligned to it. With so few accepted candidates, En→Ko MU2 is only 1.3 s faster than SEM_END, and its output equals the utterance-end output in 52/150 utterances (32/150 for Ko→En).

Extension to En↔De. Block C of Table 18 repeats the comparison on the main pair with human references: CST-Bench-TAXI turns with gold transcripts and forced-aligned word times, Qwen3.8-27B as alignment teacher and translator, $\theta = 50$, a German minimum unit set on held-out data, and trimming of committed text that is re-emitted at the turn end; `cstbench eval` scores the output against the TAXI references. Its MU2 row is the oracle-transcript streaming bound of Table 3. TAXI turns are much shorter than the pilot utterances (mean source turn 3.086 s in German and 4.624 s in English; mean pilot utterance 11.09 s in Korean and 15.05 s in English), which bounds what commits inside a turn can gain. [Draft: planned; Block C has not been run]

Limitations. The references are silver translations by one LLM, with no human references, COMET or human judgement. The utterances are long, come from single-speaker monolingual corpora and are not dialogues. Transcripts and word times are gold, latency ignores computation and ASR delay, and LAAL is computed per utterance rather than as session-level StreamLAAL. There are no confidence intervals: with 150 utterances per direction, small differences, such as between SEM_END and the utterance end, are not interpreted. The translator, the alignment teacher and the target of the generation check are the same model (Qwen3.8-27B), so the check may favour its own phrasing, and neither θ nor the minimum unit was tuned. Finally, MU2 sometimes re-emits committed text at the utterance end; the pilot does not trim it, which may inflate the Ko→En length ratio of MU2 (1.13, against 1.05 for the utterance end).

H. Statistics, Licences and Reproducibility

CST-Bench-TAXI has 86 sessions and about 3.3k reference words per direction, so every result on CST-Bench comes with an uncertainty estimate, and the statistical protocol below is fixed before any system is run on the test sets. [Draft: planned: the v0.1 scorer reports no intervals yet; the bootstrap, the Holm correction and the minimum detectable differences are being added]

Session-clustered paired bootstrap. We resample sessions rather than turns, because the turns of one session share speakers, topic and context, and resampling them independently would understate the variance. Each of 2,000 resamples draws the sessions of a test set with replacement and keeps every turn of a drawn session. Every metric is recomputed on each resample exactly as on the full test set: BLEU and chrF from the statistics pooled over the resampled turns, COMET over the resampled turns, latency statistics over the resampled non-empty turns, and error rates from pooled counts; per-session scores are never averaged. A 95% interval spans the 2.5th to the 97.5th percentile of the resampled values. A paired comparison scores both systems on the same resamples (Koehn, 2004), and its two-sided p -value is twice

the smaller fraction of resampled differences on either side of zero. On CST-Bench-Syn, the resampling unit is the source dialogue, so both speaker–language assignments of a dialogue are drawn together.

Main comparisons and multiple testing. Before the test runs, we fix the main comparisons, the operating point of every system (chosen on the CST-Bench-Syn calibration split; Appendix D) and the decision rules. The main comparisons are those behind RQ1 and RQ3 (Section 1): oracle against realistic segmentation for each system and direction, and MODEL against the same-backbone and size-matched cascades (Table 6). Holm’s step-down procedure controls the family-wise error rate over them at 0.05. Every other difference is reported with its interval but is not tested.

Rank agreement. Kendall’s τ compares system rankings between CST-Bench-TAXI and CST-Bench-Syn, between the two TTS editions, between gold and silver references, and between metrics (Table 11). Its 95% interval comes from the same bootstrap: sessions of different test sets are drawn independently, two metrics on one test set share each draw, and τ is recomputed on every resample. The decision rule for the agreement between CST-Bench-Syn and CST-Bench-TAXI is fixed in advance, and a low τ is reported as a finding.

Minimum detectable difference. For each metric and direction, we compute the smallest paired difference that the test detects at a two-sided level of 0.05 with a power of 0.8, $MDD = (z_{0.975} + z_{0.8}) \hat{\sigma}_{\Delta} \approx 2.80 \hat{\sigma}_{\Delta}$, where $\hat{\sigma}_{\Delta}$ is the session-bootstrap standard error of a paired difference, estimated from the variance of CST-Bench-TAXI segment scores. On CST-Bench-TAXI, the MDDs for En→De and De→En are [TBD] and [TBD] COMET points, [TBD] and [TBD] chrF points, and [TBD] and [TBD] s of median EndOffset. We do not interpret smaller differences and use the same estimate to size CST-Bench-Syn (Appendix B).

Release. We release one Apache 2.0 package with the CST-Bench-TAXI build scripts, frozen configurations and checksums, the session renderer, `cstbench eval` and the oracle and realistic wrappers of every baseline (Appendix D), together with the per-turn scores of every system keyed by turn ID. TAXI itself is never redistributed. Users obtain the corpus from BAS under its licence (Bavarian Archive for Speech Signals (BAS), 2001; Reichel et al., 2016) and rebuild the track deterministically; Appendix A gives the commands and the verification of timeline digests and audio hashes, and a local rebuild matched 86/86 timeline digests and audio hashes in each L0 configuration. The per-turn scores let anyone who holds the corpus compare new systems with ours without access to our outputs. No TAXI audio, transcript, translation or system output appears in the release or in this paper; apart from one KsponSpeech transcript (Figure 11), all text examples are invented. CST-Bench-Syn is released as two subsets under the licences of their sources, with a data card (Appendix B). The MODEL weights carry both component licences: OpenMDW-1.1 for the encoder and Apache 2.0 for the thinker. [Draft: planned: the realistic wrappers, CST-Bench-Syn and the MODEL weights do not exist yet]

Exact configurations. Every baseline is pinned to a public checkpoint and version (Table 12); the training stages, data and hyperparameters of MODEL are listed in Table 16; render seeds derive from session IDs (Appendix A); and computation-aware latency is measured on one fixed hardware configuration (Table 14). We report sacreBLEU signatures (Post, 2018) and the COMET checkpoint (wmt22-comet-da; Rei et al., 2022a), and we pin every package that changes scores, including the number normaliser. For review, an anonymized copy of the code is available at [TBD]. [Draft: planned; pin the number normaliser and add the anonymized link]

Licences. Table 19 lists the licence of every dataset, model and tool we use, how we use it and what we release. Three constraints shape the release: TAXI may not be redistributed; the two CST-Bench-Syn sources carry incompatible Share-Alike licences, hence two subsets; and the EXAONE AI Model License restricts EXAONE-4.0 outputs to non-commercial use, so we use them only internally (the agreement check of the CST-Bench-Syn Korean text and the pilot of Appendix G) and neither release them nor train on them. The commercial CST-Bench-Syn edition is synthesized only on a paid Azure resource.

Table 19. Licences and terms of the data, models and tools we use, how we use them, and what we release. Baseline systems and their licences are listed in Table 12. *Planned: CST-Bench-Syn and the MODEL weights do not exist yet.

Asset	Licence / terms	How we use it	Released by us
BAS TAXI 2.5 (Bavarian Archive for Speech Signals (BAS), 2001; Reichel et al., 2016)	Scientific licence, free of charge (commercial licence available); no redistribution, even in part, without the written permission of the copyright holders	CST-Bench-TAXI, the real test track; evaluation only	Session and turn IDs, configurations, build scripts, checksums, per-turn scores keyed by turn ID; no audio, text or system output
XDailyDialog (Liu et al., 2023)	CC BY-NC-SA 4.0	CST-Bench-Syn dialogues and English–German references; evaluation only	XDailyDialog-based CST-Bench-Syn subset under CC BY-NC-SA 4.0*
BConTrasT (Farajian et al., 2020), built on Taskmaster-1	CC BY-SA 4.0 (Taskmaster-1: CC BY 4.0)	CST-Bench-Syn dialogues and English–German references; evaluation only	BConTrasT-based CST-Bench-Syn subset under CC BY-SA 4.0*
Nemotron 3.5 ASR Streaming 0.6B (NVIDIA, 2026)	OpenMDW-1.1	Streaming encoder of MODEL, fine-tuned	Encoder within the MODEL weights, under OpenMDW-1.1*
Qwen3-ASR (Shi et al., 2026)	Apache 2.0	Thinker LM of MODEL (1.7B; its audio encoder removed); Qwen3-ASR-1.7B in cascade baselines and for CST-Bench-Syn re-recognition	Thinker within the MODEL weights, under Apache 2.0*
Qwen3.8-27B	Apache 2.0	MT in cascade baselines; alignment, translation and verification teacher of the MU2 commit targets; one of three translators of the CST-Bench-Syn Korean text	Per-turn scores of its runs; none of its translations as references
MADLAD-400-10B-MT	Apache 2.0	One of three translators of the CST-Bench-Syn Korean text	Korean text of CST-Bench-Syn *
EXAONE-4.0	EXAONE AI Model License: non-commercial; outputs may not be used to develop competing models	Internal evaluation only: agreement check of the CST-Bench-Syn Korean text; references and second alignment teacher in the pilot of Appendix G	Nothing; its outputs are neither released nor used for training
Azure neural TTS (Microsoft, 2026)	Service terms of the provider; synthesis on a paid pay-as-you-go resource only, as statements on the free tier conflict	Commercial CST-Bench-Syn edition (main results)	Audio of this edition*
CosyVoice 3 (Du et al., 2025) or Qwen3-TTS (Hu et al., 2026)	Apache 2.0 (weights)	Open CST-Bench-Syn edition on a subset (rank agreement)	Audio of this edition*
cst-bench code and cstbench eval	Apache 2.0	Building CST-Bench-TAXI and CST-Bench-Syn; scoring every system	All code, configurations, checksums and wrappers

[Draft: verify every entry, the TAXI copyright holders and the Azure output terms against the current licence texts before submission]